


Degree Project in Computer Science and Engineering

Second cycle, 30 credits

# Balancing Adaptation and Forgetting in Tabular Foundation Models under Concept Drift

A Study of Context Memory

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## A Study of Context Memory

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Date: November 13, 2026

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Swedish subtitle: En studie av kontextminne

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## ABSTRACT

Tabular foundation models such as TabPFN learn from labelled examples in their context, without weight updates; on a drifting stream (concept drift), the bounded context is a frozen model's only memory, and earlier situations can return. In a study of a frozen TabPFN on four public streams, pre-registered with documented amendments, we score retention (accuracy on held-out data of earlier periods, or regimes) next to adaptation (accuracy on incoming rows). Forgetting is large but hidden by stream accuracy: four context memories differ by at most 4.5 percentage points in adaptation but by up to 13.3 in retention, scored on fixed segments. It comes from evicting the oldest rows at two levels: an existing class-balanced memory does so within each class and on two streams keeps almost only the newest rows; our first regime-balanced memory does so within each regime, keeping only the end of each. Keeping old rows is what matters: three memories that do so retained 2.2 to 13.4 points more than the class-balanced one on three streams and about as much on Insects (abrupt); how they shared them among regimes mattered less.

## SAMMANFATTNING

Tabulära grundmodeller (foundation models) som TabPFN lär sig av märkta exempel i sin kontext, utan att vikterna uppdateras; på en dataström med begreppsdrift är den begränsade kontexten en fryst modells enda minne, och tidigare situationer kan återkomma. I en studie av en fryst TabPFN på fyra öppet tillgängliga dataströmmar, förregistrerad med dokumenterade ändringar, mäter vi bibehållen träffsäkerhet (retention: träffsäkerheten på undanhållna testdata från tidigare perioder, så kallade regimer) vid sidan av anpassning (träffsäkerheten på inkommande rader). Glömskan är stor men döljs av träffsäkerheten på strömmen: fyra kontextminnen skiljer sig med högst 4,5 procentenheter i anpassning men med upp till 13,3 i bibehållen träffsäkerhet, mätt på fasta segment. Den beror på att de äldsta raderna tas bort, på två nivåer: ett befintligt klassbalanserat minne gör det inom varje klass och behåller på två strömmar nästan bara de nyaste raderna; vårt första regimbalanserade minne gör det inom varje regim och behåller bara regimernas slut. Det avgörande är att gamla rader behålls: tre minnen som gör det hade 2,2 till 13,4 procentenheter högre bibehållen träffsäkerhet än det klassbalanserade minnet på tre strömmar och ungefär lika hög på Insects (abrupt); hur raderna fördelades mellan regimerna spelade mindre roll.



Keywords: Concept drift, Tabular foundation models, In-context learning, Continual learning, Catastrophic forgetting, Data streams

Nyckelord: Begreppsdrift, Tabulära grundmodeller, Kontextinlärning (in-context learning), Kontinuerlig inlärning, Katastrofal glömska, Dataströmmar

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## 1 INTRODUCTION

Tabular foundation models predict new rows from labelled example rows without any training, and they are now being applied to data that change over time. Their weights stay fixed after pre-training, so such a model learns a particular stream only from the rows it is currently shown, and only a limited number fit. Which rows should it keep? Studying a frozen TabPFN on four streams, this thesis finds that forgetting is large but invisible in accuracy on the stream; that it comes from evicting the oldest rows, within each class in an existing memory and within each regime in our own first design; and that keeping old rows is what reduces it, by 2.2 to 13.4 percentage points over that existing memory in accuracy on earlier periods, scored on fixed segments, on three of four streams but hardly at all on the fourth (Insects (abrupt)), while how the old rows are shared among regimes matters less.

### 1.1 Background: Learning from Data That Change

Many tables are recorded over time, such as gas-sensor readings or electricity prices, and the relation between the inputs and the correct label can change, so that the same input calls for a different answer than before. This is *concept drift* [12, 36]. We call a stretch of a stream in which the relation stays the same a *regime*, and the point where one regime ends a *change point*. A change can be abrupt, gradual or incremental, and earlier regimes can recur [12, 36, 50]; Section 2.1 illustrates the four types.

A learner on a stream must adapt quickly to the current regime, yet keep what it learned about earlier ones, which may be needed again. This tension between *plasticity* (taking in the new) and *stability* (keeping the old) is the stability–plasticity dilemma, studied since Grossberg [16] [see also 39].

Tabular foundation models change where this tension lies. A model such as TabPFN is pre-trained once, on a very large number of synthetic prediction tasks, and then used with frozen weights [15, 20, 21, 43]. For a new task it receives labelled rows, its *context*, and predicts new rows in one forward pass. This is *in-context learning*: like a student shown worked examples before an exercise, the model learns the task from the examples, without training. On a stream, the weights hold only the general prior from pre-training, so *the context is the model's only memory of the stream*, and a row pushed out of it is lost for good, as in instance-based stream learners [32]. For large tabular models, "the memory population effectively is the model" [34]: choosing the context rows decides the balance between stability and plasticity [34, 44], and the challenge shifts from updating the model to managing its context [27, 53].

The context cannot grow without limit, because the cost of a prediction grows with its size [20]. A stream is therefore served with a fixed *context budget*, in which recent rows favour adaptation and older rows memory of the past.

### 1.2 Problem: What Should a Bounded Context Keep?

We describe a model on a stream by two quantities (Section 3.2). *Adaptation* (ADAPT) is its accuracy on the incoming rows, measured *prequentially*: each row is predicted before its label is revealed and only then used for learning [11]. *Retention* (RET) is the accuracy that the current context gives on *holdout sets*, rows of each earlier regime kept out of the stream for this test only. All accuracies are *balanced*, the mean of the per-class accuracies [3], so that guessing the most common class scores poorly; differences are in *percentage points* (pp).

Figure 1 shows the four memories compared and, in grey, our first design. A *sliding window* keeps the most recent rows. A *dual memory* splits the budget into a *short-term part* of recent rows and a *long-term part* that receives the rows leaving it; when the long-term part is full, a row is removed (*evicted*). The

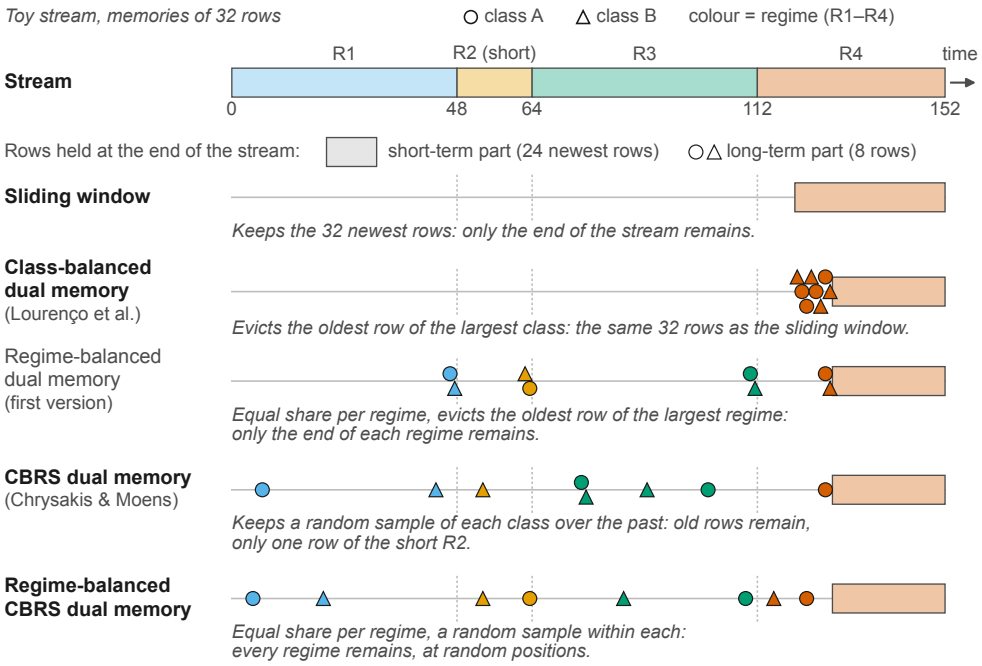



**Figure 1: Which rows each memory keeps at the end of a toy stream (schematic; one simulated run of the memory rules used in the experiments, typical of 500 runs). The stream passes through four regimes, R1–R4 (colour), of which R2 is short, with two equally frequent classes (shape) mixed in every regime. Each memory holds 32 rows, drawn at their positions in the stream: the shaded band is the short-term part (the 24 newest rows; for the sliding window all 32), the markers are the 8 rows of the long-term part. The class-balanced dual memory keeps the same rows as the sliding window, only the newest; the first regime-balanced memory (grey label, Section 4.2) keeps two rows of every regime, both from its end; the CBRS dual memory keeps a sample of the whole past but only one row of the short R2; the regime-balanced CBRS dual memory keeps two rows of every regime, spread over it.**

class-balanced dual memory (Lourenço et al. [35]) evicts the oldest row of the class with most rows. When the classes arrive mixed, without long runs of a single class, this evicts almost in order of age, and only the latest regime survives. We call this *recency collapse*; in the strict sense, a retention within 0.6 pp of a sliding window of the same size on fixed 2,000-row segments of the stream (0.3 pp when scored by regime), we observe it on two of our four streams, Gas and Electricity (Section 4.2). The CBRS dual memory (after Chrysakis and Moens [5]) keeps instead, by class-balancing reservoir sampling (CBRS), a random sample of each class over the whole stream, and the regime-balanced CBRS dual memory, our design, gives every regime an equal share of the long-term part and samples each share in the same way.

Why remember the past? Earlier situations come back, as seasons do, and must then be predicted before new labels arrive, from the old rows still in the context; RET measures this moment. This matters to those who deploy tabular foundation models on changing data, as in sensor, energy or environmental monitoring, since a model that has silently forgotten an earlier situation can fail exactly when it returns. It also matters to those who evaluate stream learners, since accuracy on the stream scores only the current regime and so hides forgetting: on Gas, two dual memories that keep either the newest rows of each class (class-balanced dual memory) or a random sample of them (CBRS dual memory) differ by 0.18 pp in

ADAPT but by 11.22 pp in retention (200 rows, fixed segments). Yet the class-balanced dual memory, proposed for TabPFN on streams [35], has been compared with other policies by prequential measures only [53]. This thesis therefore asks: *given a fixed context budget, which rows should a frozen tabular foundation model keep so that it adapts to the current regime without forgetting earlier ones, and how can such forgetting be measured?*

### 1.3 Research Questions and Contributions

**RQ1** How much does a frozen tabular foundation model forget earlier regimes of a concept-drift stream under different context memories, does the measured forgetting depend on scoring by regime, and how does it relate to the accuracy on the incoming stream?

**RQ2** Why does an existing class-balanced context memory forget earlier regimes, and does the same mechanism appear in a memory balanced by regime?

**RQ3** Which matters more for retention: keeping old rows at all, or how the long-term part is shared among regimes, and at what cost in adaptation?

The thesis makes three contributions.

(1) *Making forgetting visible.* We measure retention next to adaptation, as streaming continual learning does [6], in a study pre-registered with documented amendments: the four memories differ by up to 13.3 pp in retention on fixed segments but by at most 4.5 pp in ADAPT. *New here:* this measurement for a frozen tabular foundation model with a fixed context budget on drift streams; the studies of such models on streams that we found report only prequential or online metrics [7, 27, 35, 53, 54, 57], and the nearest study of forgetting uses image task sequences and a memory that only grows [44].

(2) *Explaining why memories forget: oldest-first eviction, at two levels.* The class-balanced dual memory evicts the oldest row within a class, so on Gas and Electricity, where the classes arrive mixed, it keeps almost only the most recent rows (94.7–97.5 % of its rows are those a sliding window of the same size would hold). Our first design, the regime-balanced dual memory, evicts the oldest row within a regime and so keeps only the last stretch of each regime (Figure 6). *New here:* measuring this failure, recency collapse, and tracing it to the eviction rule.

(3) *Keeping old rows matters most; how they are shared matters less.* All three memories that keep old rows (the CBRS dual memory, the regime-balanced CBRS dual memory and our first version) retain, on fixed segments, 2.2 to 13.4 pp more than the class-balanced dual memory on Gas, Electricity and Insects (gradual), but only $-1.1$ to $+1.3$ pp on Insects (abrupt). Equal shares per regime (the regime-balanced CBRS dual memory) retain more than the CBRS dual memory on Gas, where the regimes differ much in length, as much on Electricity, and less on the two Insects streams. Keeping old rows costs adaptation where the target depends on the recent past, up to 4.45 pp on Electricity at 200 rows. *New here:* not the principles, which come from replay memories in continual learning [4, 5, 31, 46], but their controlled comparison for the context of a frozen model, with drift regimes as the groups.

### 1.4 Scope and Structure

We study TabPFN because it is accurate on small tables without tuning [21], never trained during use, keeps a memory of rows that can be inspected one by one, and is used by the related studies on streams [7, 27, 35, 53]. We use one frozen backbone (the TabPFN-2.5 weights), classification only, four streams, contexts of 200 and 1000 rows and batches of 50 rows. The two regime-balanced memories are

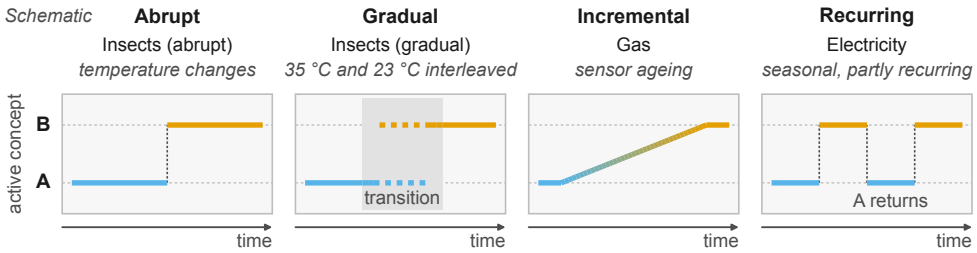



**Figure 2: Four types of concept drift (schematic). The vertical axis shows which concept generates the data. Abrupt: the concept changes at one point in time. Gradual: during a transition, instances of the old and the new concept are interleaved, and the new one becomes more frequent. Incremental: the concept moves step by step from A to B. Recurring: an earlier concept returns. The line under each panel title names the stream of this study that has this type of drift (Table 3).**

given the documented regime boundaries; a variant of the first that detects them is run on all four streams. Stream learners that are not foundation models are not compared.

Section 2 reviews related work, Section 3 describes the method, Section 4 answers RQ1–RQ3, Section 5 discusses the results, including sustainability and ethics, and Section 6 concludes.

## 2 BACKGROUND AND RELATED WORK

For each of five lines of work, this section states its usual approach, why it leaves our question open, and what our design takes from it (Table 1); we found no study that measures what a frozen tabular model with a bounded context forgets on a drifting stream.

### 2.1 Concept Drift and Learning from Data Streams

In stream learning, each arriving row is predicted before its label is known, and the data drift in the ways of Figure 2. The common approach reacts to drift: detect the change, then forget outdated data or models. ADWIN keeps a window of recent observations whose length adapts to the rate of change [2]; adaptive random forests replace a tree when its detector fires [13]. Windowing assumes that the most recent data are the relevant ones, which fails when concepts recur [12]; recurring concepts are handled by storing whole models and reusing them [10, 14, 24]. Evaluation is prequential (test-then-train), often with fading factors that deliberately forget old errors [11]; the INSECTS benchmark, the source of two of our streams, is evaluated prequentially [50].

Instance-based learners, whose model is the set of stored examples, are the closest analogue to a frozen foundation model: SAM-kNN keeps a short-term memory for the current concept and a long-term memory of older, non-contradicting examples for recurring ones [32, 33], an analogy drawn by Verma et al. [53]. Streaming continual learning joins the rapid adaptation of stream learning with the retention sought by continual learning [6, 17]: Cossu et al. [6] score prequential Cohen's kappa (accuracy corrected for chance) and, after each drift, kappa on held-out sets of all earlier concepts, but for learners that update parameters. In other stream-learning work, accuracy on earlier concepts is not a standard measure [12, 36]. We adopt the two-sided protocol for a model whose memory is a fixed-size context.

### 2.2 Continual Learning: Stability, Plasticity and Replay

Continual learning studies models that learn a sequence of tasks, framed as the stability–plasticity dilemma [16, 39, 45]. Its failure mode is *catastrophic forgetting*: training on new data overwrites the weights that encoded earlier knowledge [9, 38]. Remedies protect important weights [25] or replay examples from a small memory, and forgetting is measured as the drop in accuracy on earlier tasks (backward transfer) [31]. Replay memories are balanced by class [5, 46, 48], by task [31] or by both [4]; replay designed for drift tends to remove drifted examples rather than protect them [1, 26]; prompt-based methods freeze the backbone but learn small prompts [55, 56].

All of these update parameters, so the question of what the memory alone lets a frozen model retain does not arise. We borrow two choices from them and apply them to the context: how the memory is shared among groups, here classes or drift regimes, and which row of a group is evicted. Class-balancing reservoir sampling evicts a random row of the largest class and GDumb refuses new rows of full classes, so neither favours recent rows [5, 46]; evicting the *oldest* row of the largest class, as the class-balanced dual memory does, can. For replay, a reservoir over the whole stream is a strong default, and memories balanced by group help mainly when each group gets only a few slots [4].

### 2.3 Tabular Foundation Models and In-Context Learning

A prior-data fitted network is a Transformer trained once on many synthetic datasets so that one forward pass approximates Bayesian prediction for a new dataset [43]. TabPFN applies this to tables [15, 20, 21]; TabICL is a related model for larger tables [47]. Without positional encoding, row order carries no information [43], so what matters is which rows the context holds. Context selection targets large static datasets: retrieving neighbours of each query [51], routing queries to data subsets [58] or tuning a compressed context [8]. Temporal shift has been addressed through the pre-training prior [19], and the TabPFN authors list data drift as future work [21]. What a bounded context should keep on a drifting stream is open; we keep the model fixed and treat its context as a time-ordered memory with a budget chosen for the stream.

For language models the prompt likewise carries task knowledge to a frozen model. Studies ask which demonstrations to retrieve [29, 49], what they convey [40] and how order and position bias the answer [30, 37, 59]; others measure forgetting when a frozen language model learns tasks or classes in sequence through its prompt [23, 41], also in theory [28]; there, old classes keep their meaning and nothing is evicted under a budget. The recency bias of language models lies in the *reader*, which weights the end of a prompt [59], whereas recency collapse lies in the *memory policy*.

### 2.4 Tabular Foundation Models on Data Streams

Lourenço et al. [35] combine a short-term FIFO queue (first in, first out) with a long-term store balanced over classes, meant to keep representative samples of recurring classes. Evaluated prequentially only, the dual memory is on average 0.31 pp more accurate than the short-term memory alone, with no significant difference on five of seven streams [35]. Its authors call the design naive, its stability addressing forgetting of classes only [34]. CURE admits rows by the model's uncertainty and evicts a redundant row of the most represented class [27]; HINT keeps a sliding window on an edge device and sends uncertain rows to TabPFN in the cloud [7]. Wu et al. [57] store a context of a frozen TabPFN for each detected concept, each with its last 1,000 rows, and mix them as experts; they compare with FIFO and the class-balanced dual memory and test accuracy in the batches right after a concept returns, but do not score held-out data

**Table 1: The thesis compared with the five lines of related work.**

| Line of work | Learning happens in | What is kept | Evaluation | Retention of earlier regimes measured? |
|---|---|---|---|---|
| Stream learning (Section 2.1) | parameters; stored examples in instance-based learners | recent window, pool of models; short- and long-term example memory (SAM-kNN) | prequential | mostly no; yes in streaming continual learning, for parameter-based learners |
| Continual learning and replay (Section 2.2) | parameters | replay memory balanced by class, task, or task and class | accuracy on earlier tasks, backward transfer | yes, on task sequences |
| Tabular foundation models (Section 2.3) | context (frozen weights) | one static table | static benchmarks | no |
| In-context learning in language models (Section 2.3) | context (frozen weights) | one prompt; sequences of tasks or classes | per-query accuracy; old-task accuracy | partly; task or class sequences without drift |
| Tabular foundation models on streams (Section 2.4) | context (frozen weights) | bounded rows: FIFO, class-balanced dual memory, CURE, retrieval, contexts per concept | prequential or online; accuracy right after a concept returns | no (one image-task study with a growing memory) |
| **This thesis** | context (frozen TabPFN) | fixed budget; long-term part shared by class or by regime, oldest-first or reservoir | ADAPT and RET | yes, on tabular concept-drift streams |

of earlier regimes. Wang et al. [54] correct the predictions of a frozen in-context tabular model under temporal drift with a nearest-neighbour residual corrector in its embedding space, fed by past labelled rows; on eight industrial streams they attribute the gains to four mechanisms and characterise theoretically what such correction can and cannot recover. Their adaptation acts on the outputs rather than the context rows and is scored on the current error, not on earlier regimes. Narayanan et al. [44] measure forgetting of a frozen TabPFN on image task sequences with a memory that only grows.

The closest work, Verma et al. [53], compares memory policies across tabular foundation models and finds keeping only the most recent rows as effective as the others, including the class-balanced dual memory; memory size matters far more than the policy (their Tables 2 and 3). On their one stream with recurring concepts, TabPFN scores 0.646 with both a FIFO and the dual memory (their Table 9), which fits either retention not helping or the dual memory not retaining; prequential accuracy cannot tell. They analyse memory content only analytically, for a full FIFO memory (their Appendix F.1); what a dual memory keeps is not measured, and no policy allocates the context by regime.

## 2.5 Positioning of This Thesis

Table 1 reads row by row: stream learning mostly scores the current concept, continual learning protects parameters, and work on tabular foundation models for streams reports only prequential or online metrics.[1] The three gaps lead to three design decisions: we measure adaptation and retention together (Section 3.2), we measure what the stores actually keep (RQ2), and we separate keeping old rows from sharing them among regimes (RQ3, Section 3.3).

[1]The literature search was made in October and November 2026 in OpenAlex, arXiv, Semantic Scholar, Crossref, DataCite and PMLR; Google Scholar, DBLP, the ACM Digital Library and OpenReview were not searched directly. Every reference was checked in two independent sources.

# 3 METHOD

## 3.1 Research Process and Change of Research Direction

The project set out to build a three-level system around a frozen TabPFN: a fast corrector of its outputs, a gated adapter adding a small learned correction per regime, and a drift detector whose alarms trigger actions such as resetting the context. On synthetic streams it helped on one of three streams and harmed on another; on Insects, the ADWIN detector [2] of the river library [42] raised alarms, but no action after an alarm gave a measurable gain, even at the documented change points, because few errors after a change could be recovered, and the system made more errors than the frozen model alone (Appendix A). In the analysis of Wang et al. [54], correcting a frozen tabular model's outputs from past labelled rows helps on seven of eight temporal streams but harms on one; our negative results fit this stream dependence, although the two correctors differ in design.

A frozen model already adapts through its context, so the question became what the context should keep, and how much the model forgets when it keeps the wrong rows. The new study was *pre-registered*: question, data, methods, metrics and hypotheses were fixed in writing before the main experiments, and every later change was added as a dated amendment with its reason. The method registered at the outset, an archive of rows per finished regime, failed its pilot criterion by losing too much accuracy on the stream, and the pilot showed the class-balanced dual memory retaining little of an early regime; an amendment therefore replaced it by the regime-balanced dual memory before the main runs (Appendix C).

A review of the first draft asked whether the gains of the regime-balanced dual memory came from sharing the memory by regime or from knowing the regime boundaries, which the official retention measure also uses. An amendment registered before any new run added four analyses: the boundary-free variant on every stream, a long-term part that needs no boundaries (the CBRS dual memory), the regime index as an extra feature, and retention on fixed segments, which needs no new runs and had already been computed from the stored predictions, as the amendment discloses. Their results, and a later replay of the memories from the labels, showed that the regime-balanced dual memory keeps only the newest rows of each regime; the regime-balanced CBRS dual memory, designed from this finding, was registered before its runs. These later rounds, 5 and 6 (Section 3.5), are exploratory.

## 3.2 Learning Setting and Evaluation Protocol

As Figure 3 shows, from row 200 of a time-ordered stream on, rows are processed in batches of 50 that never span a regime boundary: the model predicts a batch from a context of earlier rows only, the labels are revealed, and the memory policy takes in the new rows. Features are standardised with the statistics of the first 200 rows. Before the stream starts, a holdout set is drawn from every regime, stratified by class, and removed from the stream; at every *checkpoint* the current memory predicts the holdouts of all regimes seen so far. Run-time checks ensure that checkpoints never change the memory and that the context never holds a future row or exceeds its budget.

ADAPT is the balanced accuracy over all predicted stream rows (random guessing gives $1/K$ for $K$ classes). RET is, for each ended regime, the balanced accuracy on its holdout set averaged over the checkpoints after its end, then averaged over these regimes. In plain terms, RET asks: if an earlier regime came back now, before any of its new labels arrived, how well would the current context do?

RET averages over the same regimes that the regime-balanced memories use, so a different cut, for example months instead of seasons on Electricity, would change these memories and RET together. To score retention on a unit that no memory uses, $\text{RET}_{\text{seg}}$ cuts the stream into fixed segments of 2,000 rows,

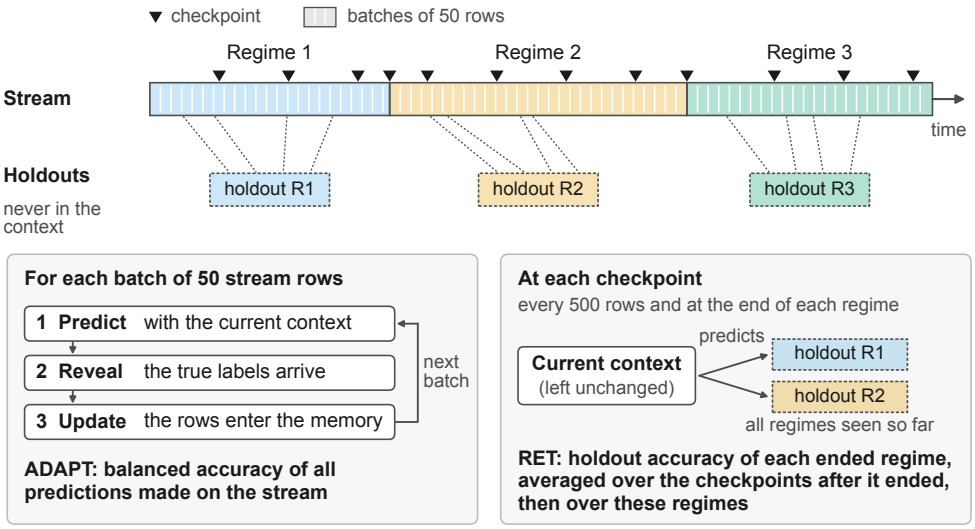



**Figure 3: How adaptation (ADAPT) and retention (RET) are measured. The stream is processed in batches of 50 rows: the frozen model predicts each batch from its current context, the true labels are then revealed, and the memory policy updates the context; ADAPT is the balanced accuracy of all these predictions. A holdout set drawn from each regime never enters the context; at every checkpoint (every 500 rows and at the end of each regime) the current context, left unchanged, predicts the holdouts of all regimes seen so far; RET uses those of the regimes that have ended, averaged over checkpoints and then over regimes. $\text{RET}_{\text{seg}}$ averages over fixed 2,000-row segments instead of regimes.**

scores the holdout rows of each segment at every checkpoint after its end, and averages the segments with equal weight. This length leaves at least 24, and nearly always 30 or more, holdout rows in every segment used and gives 6 (Gas) to 25 (Insects (abrupt)) segments per stream; 1,000 and 5,000 rows lead to the same conclusions (Table 11 in Appendix D). The official RET serves the comparisons registered for rounds 2 and 3, $\text{RET}_{\text{seg}}$ those added later; round 4, registered with the official RET, is reported in both. $\text{RET}_{\text{worst}}$, the retention of the worst-retained regime, was defined after the round-5 results and is reported as a post-hoc measure.

The model is TabPFN with the TabPFN-2.5 weights [15] (tabpfn 6.4.1), frozen, averaging four forward passes on slightly varied inputs. Each setting is run with ten *seeds*, which fix the holdout rows, TabPFN's random state and the random draws of the two CBRS memories; the stream order never changes. "10/10" means that a difference has the same sign in all ten runs.

### 3.3 Context Memory Policies

Each memory keeps at most $B = 200$ or $B = 1000$ rows (Figure 1). The sliding window keeps the $B$ most recent rows, which amounts to using TabPFN alone. The four dual memories share the structure of Lourenço et al. [35]: a short-term FIFO queue holds 75 % of the budget (150 or 750 rows), and its oldest row moves into a long-term part of 50 or 250 rows. They differ only in which row the long-term part evicts when it is full, which involves two choices taken from replay memories in continual learning [4, 5, 31, 46] (Table 2).

**Table 2: The four dual memories, sorted by how the long-term part is shared and which row is evicted within a group. The first row needs no regime boundaries.**

| Shared among | Evict the oldest row | Class-balancing reservoir sampling |
|---|---|---|
| classes only | Class-balanced dual memory [35] | CBRS dual memory [5] |
| regimes, then classes | Regime-balanced dual memory (first version) | Regime-balanced CBRS dual memory |

*Class-balanced dual memory.* When the long-term part is full, the oldest row of its most common class is evicted (Algorithm 1 of Lourenço et al. [35]).

*CBRS dual memory.* The long-term part follows class-balancing reservoir sampling [5]. A row of a class that has never been the largest replaces a random row of the largest class; a row of a class that has been the largest replaces a random row of its own class with probability $m_c/n_c$, the number of its rows stored over the number seen so far, and is otherwise discarded. Each class thus holds a uniform random sample of its rows over the whole stream, without any regime information.

*Regime-balanced dual memory (first version).* Every row carries the regime in which it arrived. When the long-term part is full, the memory finds the regime with most rows in it (on a tie, the newer), within it the class with most rows, and evicts the oldest row of that class in that regime. Every regime keeps a roughly equal share, but within a regime only its newest rows survive (Section 4.2). Without boundaries it reduces to the class-balanced dual memory; it is the method of rounds 2 to 4.

*Regime-balanced CBRS dual memory.* It applies the CBRS rule twice: to the regimes, as if they were classes, so that every regime receives an equal share, and to the classes within each regime, with a reservoir inside every combination of regime and class. With a single regime it follows exactly the rule of the CBRS dual memory.

*Regime boundaries.* The two regime-balanced memories receive the documented change points, as the continual-learning strategies of Cossu et al. [6] receive theirs; the other memories need none. This separates *when* the data changed (detection) from *what* to keep (allocation), and the thesis studies allocation. In many applications the boundaries are known, from events such as a new season, a new batch of sensors or a replaced device; otherwise they are not available when a model is deployed, so the runs with the documented change points assume perfect detection. A *boundary-free variant* of the regime-balanced dual memory drops this assumption: it closes a regime when ADWIN [2], run on its own errors, raises an alarm ($\delta = 0.002$, a pause of 300 rows after an alarm, regimes of at least 500 rows), on all four streams. Only its memory is free of the boundaries: as in every run, its batches of 50 rows are still cut at the documented change points, and RET still scores the documented regimes, so the variant is compared in $\text{RET}_{\text{seg}}$. A last control gives the class-balanced dual memory the regime index as an extra feature (Appendix D).

### 3.4 Data Streams

Figure 4 and Table 3 describe the four streams, chosen because their drift differs (Figure 2); in Gas the sensors age and earlier conditions do not return, while Electricity's seasons partly recur [18]. As to data ethics, all four are public research benchmarks recorded from insect sensors, gas sensors and an electricity market; none contains data about persons, and their sources, and licences where stated, are

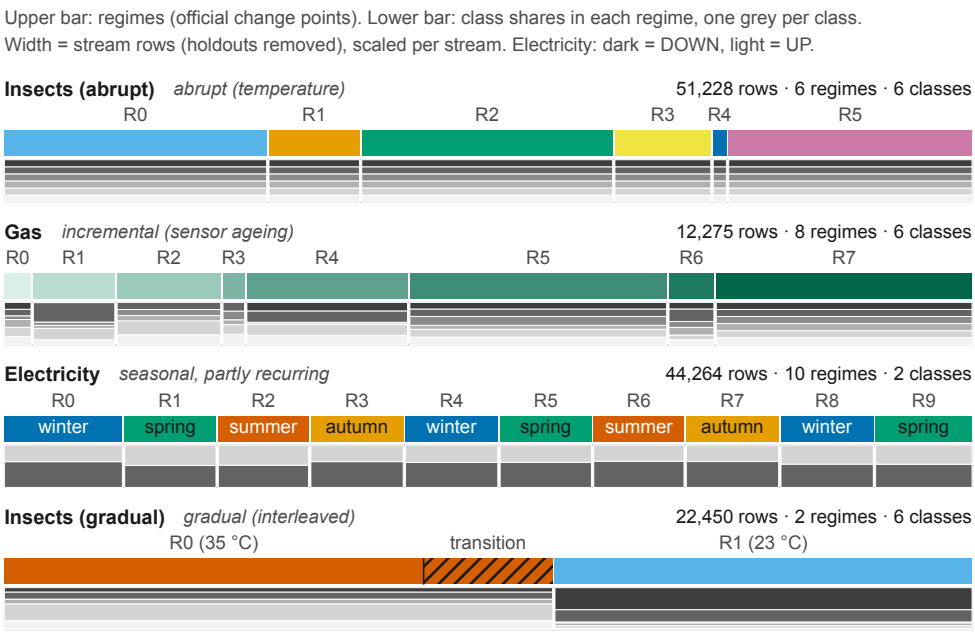



**Figure 4: The four data streams. Upper bar: the regimes, cut at the official change points; positions are stream rows after the holdout rows have been removed, and each stream is scaled to the full width. Lower bar: the share of each class within each regime (one grey per class). Class shares are equal in every regime on Insects (abrupt), but its first four change points are each preceded by a long run of a single class (Appendix B); class shares vary between regimes on Gas, stay roughly constant on Electricity, and change between R0 and R1 on Insects (gradual). On Gas and Electricity the classes are mixed within each regime, and no such run precedes a change point. Electricity's regimes are seasons that come back each year. The hatched area on Insects (gradual) is the 3,000-row transition, which belongs to R0.**

**Table 3: The four data streams (rows, classes, regimes and drift type are shown in Figure 4). Holdout rows are drawn from every regime, removed from the stream and used only to measure retention.**

| Stream | Regimes and label | Features | Holdout rows | Source |
|---|---|---|---|---|
| Insects (abrupt) | INSECTS balanced abrupt stream; regimes cut at its five documented changes of temperature | 33 | 1,620 | [50] |
| Gas | Gas Sensor Array Drift (UCI 224); ten time-ordered batches over 36 months, batches of fewer than 300 rows merged into the next (4 with 5, 8 with 9) | 128 | 1,635 | [52] |
| Electricity | New South Wales market (OpenML 151); regimes are seasons; label: price above its mean over the previous 24 hours | 4 | 1,000 | [18] |
| Insects (gradual) | 23,000-row slice of the INSECTS imbalanced incremental-gradual stream; R0 is 35 °C ending in a 3,000-row transition, R1 is 23 °C | 33 | 550 | [50] |

given in Appendix B. The three streams added to Insects (abrupt) met criteria fixed in the pre-registration; every stream that met them is used, and none was dropped after results were seen (Appendix B).

**Table 4: The rounds of the study. Runs: new runs of the round (ten seeds per setting); Tests: how the round's comparisons are tested (Appendix C).**

| Round | Compared | Streams, rows | Runs | Tests |
|---|---|---|---|---|
| 1 | Three-level system with a drift detector | synthetic, Insects, Electricity | – | preliminary study, negative (Appendix A) |
| 2 | 16 policies, among them windows of 100 to 2000 rows, the class-balanced dual memory, the regime-balanced dual memory with and without given boundaries, and regime archives | Insects (abrupt) | 160 | registered hypotheses H1–H8, primary H1–H3 |
| 3 | Window, class-balanced dual memory, regime-balanced dual memory | four streams, 200 and 1000 | 240 | registered Holm family (Amendment 3) |
| 4 | Short-term shares 0.65 and 0.85 of the two dual memories of round 3 | four streams, 200 | 160 | exploratory |
| 5 | Boundary-free variant, CBRS dual memory, regime-index control | four streams, 200 and 1000 | 210 | exploratory (Amendment 4) |
| 6 | Regime-balanced CBRS dual memory | four streams, 200 and 1000 | 80 | exploratory (Amendment 5) |

### 3.5 Experimental Design and Statistical Analysis

After the preliminary study (round 1), five rounds ran on one laptop GPU (NVIDIA RTX 2060 Max-Q, 6 GB), with ten seeds per setting (Table 4).

We report means and standard deviations (SD) over seeds. Two policies are compared through *paired differences* within each seed, so that the seed's effect cancels, with a 95 % $t$ interval, the number of positive seeds and a paired $t$-test. Within a family of tests, $p$-values are adjusted by the Holm method [22], which keeps the chance of any false positive in the family below 5 %. The text gives the sign count and Holm significance; most intervals are in Figure 7 and in Table 10 in Appendix D, the adjusted $p$-values of the round-3 family in Appendix C; all others can be recomputed from the per-seed results (Appendix B).

The hypotheses of round 2 have fixed thresholds and one-sided, Holm-corrected primary tests; several compare the 1000-row memories with the pre-registered 200-row sliding window, one fifth of the larger budget, which we call the 200-row baseline. The eight RET comparisons of round 3 between the regime-balanced dual memory and the class-balanced dual memory form a two-sided Holm family without success thresholds. In rounds 5 and 6 each registered comparison between two memories forms its own Holm family of the eight combinations of stream and budget; the other comparisons are descriptive (Appendix C). Every dual memory used in rounds 3 to 6 is also replayed without the model and must reproduce its fingerprint at every checkpoint, and unit tests cover the policies. As to research integrity, the hypotheses were fixed before the main runs and changed only by documented amendments (Appendix C), the failed hypotheses and the negative preliminary study are reported (Section 4.3, Appendix A), and code and per-seed results are public (Appendix B).

## 4 RESULTS

Two checks come first. The frozen model learns every stream: every memory adapts far better than the better of two trivial baselines that use the same revealed labels (on Electricity, where the margin is smallest, at least 76.54 against 54.01 %; Appendix B). And the runs are sound: all run-time checks and replays of the memories without the model passed (Appendix B).

### 4.1 RQ1: Forgetting Is Large, Hidden by Stream Accuracy, and Not Due to the Scoring

The frozen model forgets much of what it has seen. With a sliding window, its accuracy on earlier regimes is far below that on the stream (Table 5): 35.08 against 75.68 % on Insects (abrupt) at 200 rows, where

**Table 5: Adaptation (ADAPT) and retention of the four context memories: balanced accuracy in %, mean over ten seeds. RET: official retention, averaged over the earlier regimes; $RET_{seg}$: retention on fixed 2,000-row segments, whose boundaries do not depend on the regimes (Section 3.2). Window: sliding window; Class-bal.: class-balanced dual memory (Lourenço et al. [35]); CBRS: CBRS dual memory (after Chrysakis and Moens [5]); Reg.-bal. CBRS: regime-balanced CBRS dual memory. The highest $RET_{seg}$ of each block is in bold.**

| Stream | Size | Memory | ADAPT | RET | $RET_{seg}$ |
|---|---|---|---|---|---|
| Insects (abrupt) | 200 | Window | 75.68 | 35.08 | 40.68 |
| | | Class-bal. | 74.76 | 42.72 | 49.78 |
| | | CBRS | 74.67 | 43.34 | **49.88** |
| | | Reg.-bal. CBRS | 74.77 | 42.68 | 48.64 |
| Insects (abrupt) | 1000 | Window | 76.99 | 39.76 | 47.26 |
| | | Class-bal. | 76.65 | 45.86 | 53.89 |
| | | CBRS | 76.74 | 47.49 | **55.16** |
| | | Reg.-bal. CBRS | 76.90 | 47.11 | 54.20 |
| Gas | 200 | Window | 98.54 | 78.67 | 84.08 |
| | | Class-bal. | 98.55 | 78.85 | 84.33 |
| | | CBRS | 98.73 | 93.13 | 95.55 |
| | | Reg.-bal. CBRS | 98.70 | 96.37 | **97.39** |
| Gas | 1000 | Window | 99.02 | 81.74 | 86.50 |
| | | Class-bal. | 99.03 | 82.02 | 87.02 |
| | | CBRS | 99.11 | 96.93 | 98.05 |
| | | Reg.-bal. CBRS | 99.08 | 98.42 | **98.93** |
| Electricity | 200 | Window | 82.33 | 64.48 | 66.53 |
| | | Class-bal. | 82.58 | 64.48 | 66.53 |
| | | CBRS | 78.13 | 68.55 | 69.91 |
| | | Reg.-bal. CBRS | 79.19 | 69.11 | **70.28** |
| Electricity | 1000 | Window | 77.43 | 65.68 | 68.39 |
| | | Class-bal. | 77.68 | 65.38 | 68.14 |
| | | CBRS | 77.25 | 71.41 | **73.10** |
| | | Reg.-bal. CBRS | 77.80 | 71.15 | 72.80 |
| Insects (gradual) | 200 | Window | 68.80 | 27.52 | 47.59 |
| | | Class-bal. | 72.27 | 25.74 | 50.16 |
| | | CBRS | 70.93 | 48.16 | **55.08** |
| | | Reg.-bal. CBRS | 71.32 | 39.11 | 53.35 |
| Insects (gradual) | 1000 | Window | 74.48 | 31.89 | 53.11 |
| | | Class-bal. | 76.25 | 31.36 | 55.90 |
| | | CBRS | 75.06 | 55.04 | **61.37** |
| | | Reg.-bal. CBRS | 75.19 | 50.43 | 61.14 |

The standard deviation over the ten seeds is at most 1.31 pp for ADAPT, 3.67 pp for RET and 2.07 pp for $RET_{seg}$; Figure 7 gives paired differences with 95 % intervals. The values of every seed are in `results/round5_per_seed.csv` of the public repository (Appendix B).

random guessing gives 16.7 %. Once a regime has ended and its rows have left the window, accuracy on it drops (Figure 11 in Appendix D); with 1000 rows, accuracy on R1 even falls below chance by the end of the stream, because R1 is almost unrelated to the other regimes and their rows cannot stand in for it (Figure 15 in Appendix D).

The choice of memory matters far more for retention than for accuracy on the stream (Figure 5): within each combination of stream and budget, the four memories differ by 3.8 to 13.3 pp in $RET_{seg}$ but by 0.1 to 4.5 pp in ADAPT. Only on Electricity at 200 rows is the spread in ADAPT (4.5 pp) larger than that in $RET_{seg}$ (3.8 pp), a cost discussed in Section 5.2.

The forgetting does not depend on scoring by regime. On both RET and $RET_{seg}$, the class-balanced dual memory retains about as little as the window of the same size on Gas and Electricity, and the CBRS dual memory, which uses no regime information, retains 3.4 to 11.2 pp more than the class-balanced

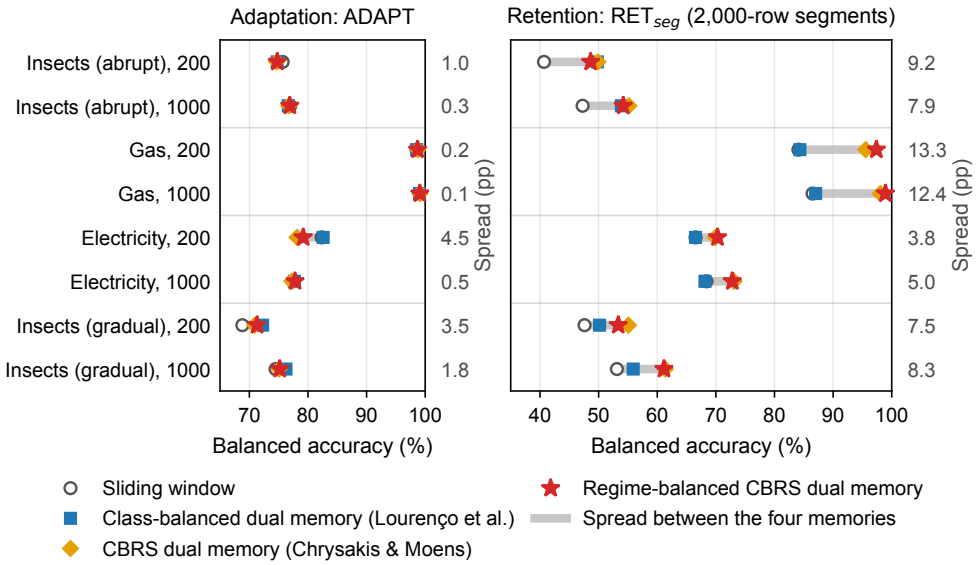



**Figure 5: Adaptation (left) and retention on fixed segments (right) of the four memories at the same context size (mean over ten seeds; numbers beside each panel: the spread between the four memories, in pp). In ADAPT the memories lie close together, in RET$_{seg}$ far apart: the choice of memory changes retention much more than accuracy on the stream. The values are in Table 5.**

dual memory on Gas, Electricity and Insects (gradual) in RET$_{seg}$, in every seed. On Insects (abrupt) the three dual memories retain about equally (Section 4.2), all far more than the window.

*Answer to RQ1.* With a sliding window, a frozen model forgets much of each earlier regime once its rows leave the context. The memory changes retention by up to 13.3 pp in RET$_{seg}$ but adaptation by at most 4.5 pp, so accuracy on the stream does not reveal forgetting; the forgetting of the class-balanced dual memory appears under both measures and next to a memory that needs no regime boundaries.

### 4.2 RQ2: Oldest-First Eviction Makes Memories Forget, at Two Levels

*First level: the class-balanced memory.* When the classes arrive mixed, without long runs of a single class, the row that the class-balanced dual memory evicts, the oldest of the largest class, is nearly always among the oldest rows overall, so the long-term part acts as a queue that extends the short-term one, and an earlier regime leaves the memory soon after it ends; at the end of the stream the long-term part holds rows of the last regime only (Figure 6; over time in Figure 12 in Appendix D). On Gas, whose class shares vary between regimes, rows of a class that later becomes rare stay for a while, but none is left at the end.

Two measures show how close this memory comes to a sliding window of the same size. On Gas and Electricity, 94.7–97.5 % of its rows are among those the window would hold, and its retention differs from the window's by only $-0.26$ to $+0.52$ pp in RET$_{seg}$ ($-0.30$ to $+0.28$ pp in RET): this is recency collapse in the strict sense. The two Insects streams differ. On Insects (abrupt) the class-balanced dual memory retains clearly more than the window ($+9.10$ and $+6.62$ pp in RET$_{seg}$, 10/10); a long run of one class precedes four of its five change points (Appendix B), and a plausible, untested explanation is that such runs leave old rows of the other classes in the long-term part. On Insects (gradual), it replaces R0 soon after the documented change point.

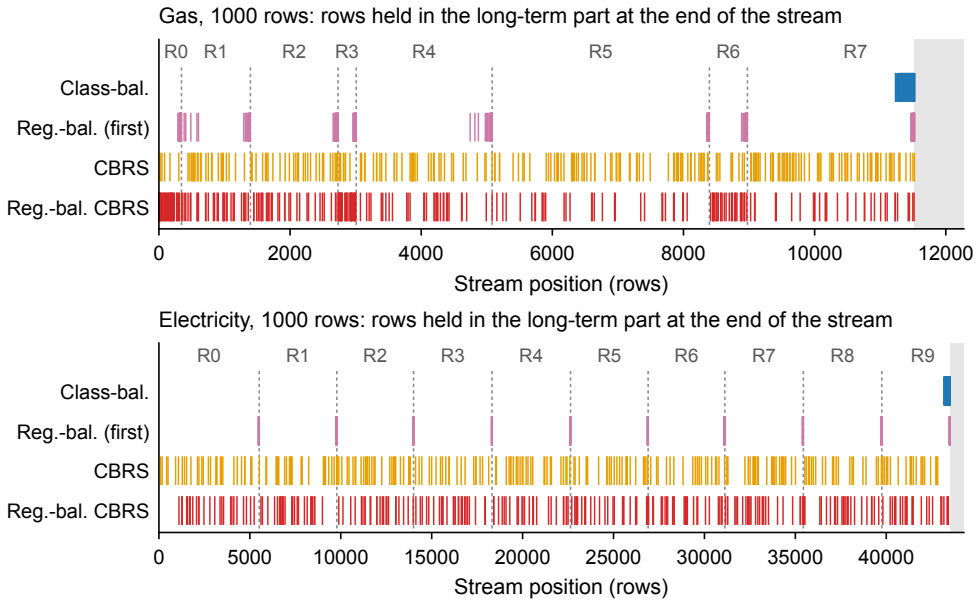



**Figure 6: Which rows the 250-row long-term part holds at the end of the stream (1000-row context, one run, seed 0); each tick is one row at its position in the stream, dashed lines are the ends of regimes, and the grey band is the short-term part, the most recent 750 rows, the same for all memories. The class-balanced dual memory holds only rows of the last regime. The regime-balanced dual memory holds 25 to 32 rows of every regime, almost all from the last stretch of that regime (a few rows of a class rare in the regime are older). The CBRS dual memory spreads its rows over the stream but in proportion to what each regime contributed (on Gas 5 rows of R0 and 6 of R3, 73 of R7). The regime-balanced CBRS dual memory holds an equal share of every regime, spread across it.**

*Second level: the first regime-balanced memory.* The regime-balanced dual memory shares the long-term part among regimes but evicts the oldest row within a regime, so after a regime ends only its last stretch survives (Figure 6): over ten seeds, its rows lie at a mean position of 0.78 to 1.00 within their regime (0 is the start, 1 the end), and on Electricity an earlier season is represented by its last one or two days only. Where a regime's holdout rows differ from its last days, this costs retention. Against the CBRS dual memory, which samples each class over the whole stream, the regime-balanced dual memory retains significantly more only on Gas at 200 rows ($+2.17$ pp, 9/10; $+0.67$ pp, 7/10, not significant, at 1000 rows), and less on Insects (abrupt) at 1000 rows ($-1.02$ pp), on Electricity at 1000 rows ($-2.47$ pp) and on Insects (gradual) ($-2.74$ and $-1.37$ pp), in $\mathrm{RET}_{\mathrm{seg}}$, with no seed in its favour and each significant after Holm correction. At 1000 rows, sampling uniformly within each regime, as the regime-balanced CBRS dual memory does, removes most of this loss on Electricity and Insects (gradual): $+2.17$ and $+1.13$ pp over the regime-balanced dual memory (10/10 and 9/10, both significant after Holm correction); in the other six combinations the difference is not significant, including $+1.00$ pp on Insects (gradual) at 200 rows.

*Answer to RQ2.* Both memories forget for the same reason: they evict the oldest row of a group. In the class-balanced dual memory the group is a class, and where the classes arrive mixed, on Gas and Electricity, it reduces to a recency window (recency collapse); in the first regime-balanced memory the group is a regime, and it keeps every regime but only its last stretch. Memories that do not evict by age,

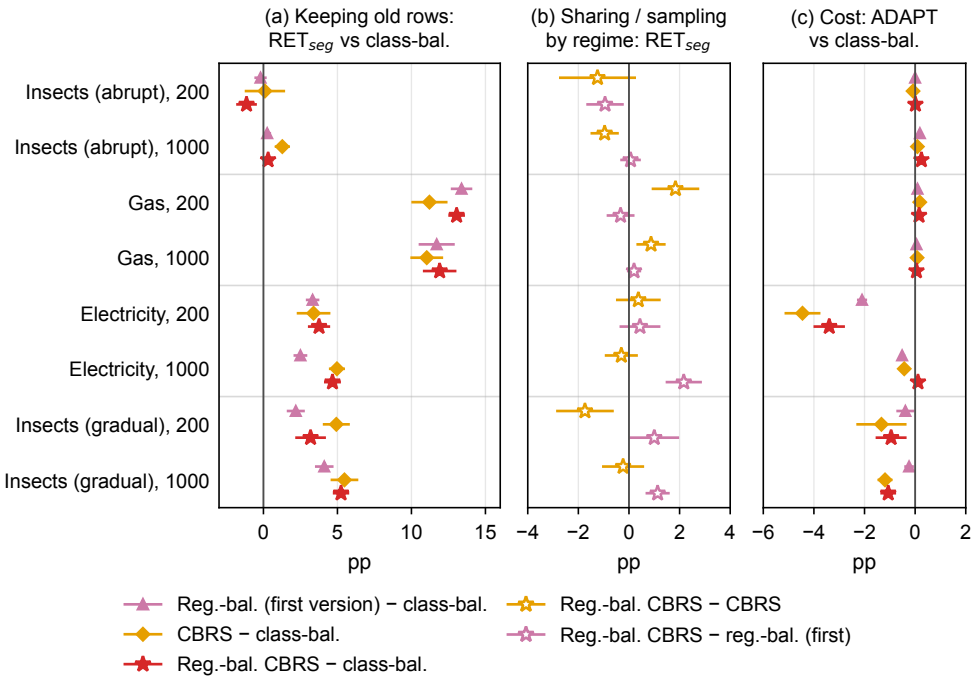



**Figure 7: Paired differences between memories at the same context size (ten seeds; markers: mean; bars: two-sided 95 % $t$ intervals). (a) Keeping old rows: RET$_{seg}$ of the three memories that keep old rows minus that of the class-balanced dual memory (Lourenço et al. [35]). (b) Sharing among regimes (regime-balanced CBRS dual memory minus CBRS dual memory) and sampling within a regime (regime-balanced CBRS dual memory minus the first version), in RET$_{seg}$. (c) Cost: ADAPT of the three memories minus that of the class-balanced dual memory. The panels use different scales.**

the CBRS dual memory and the regime-balanced CBRS dual memory, keep old rows from the whole stream.

### 4.3 RQ3: Keeping Old Rows Matters Most; How They Are Shared Matters Less

*Keeping old rows.* All three memories that keep old rows (the CBRS dual memory, the regime-balanced CBRS dual memory and the first version) retain more than the class-balanced dual memory on Gas, Electricity and Insects (gradual), by 2.2 to 13.4 pp in RET$_{seg}$ and in every seed (Figure 7a); on Insects (abrupt), where the class-balanced dual memory already keeps some old rows, the differences are small ($-1.14$ to $+1.28$ pp). The comparison registered before round 3, the regime-balanced dual memory against the class-balanced dual memory in the official RET, is significant after Holm correction in all eight cases and in all ten seeds (Appendix C): $+3.21$ to $+18.00$ pp on the other three streams, but only $+1.45$ and $+1.42$ pp on Insects (abrupt), a gain that fixed segments do not confirm ($-0.20$ and $+0.25$ pp). On the other three streams fixed segments keep the gain in every seed but make it smaller ($+2.19$ to $+13.39$ pp).

*How the rows are shared.* Between the two memories that sample old rows uniformly, the way the long-term part is shared matters much less (Figure 7b), and neither retains most on every stream. Against

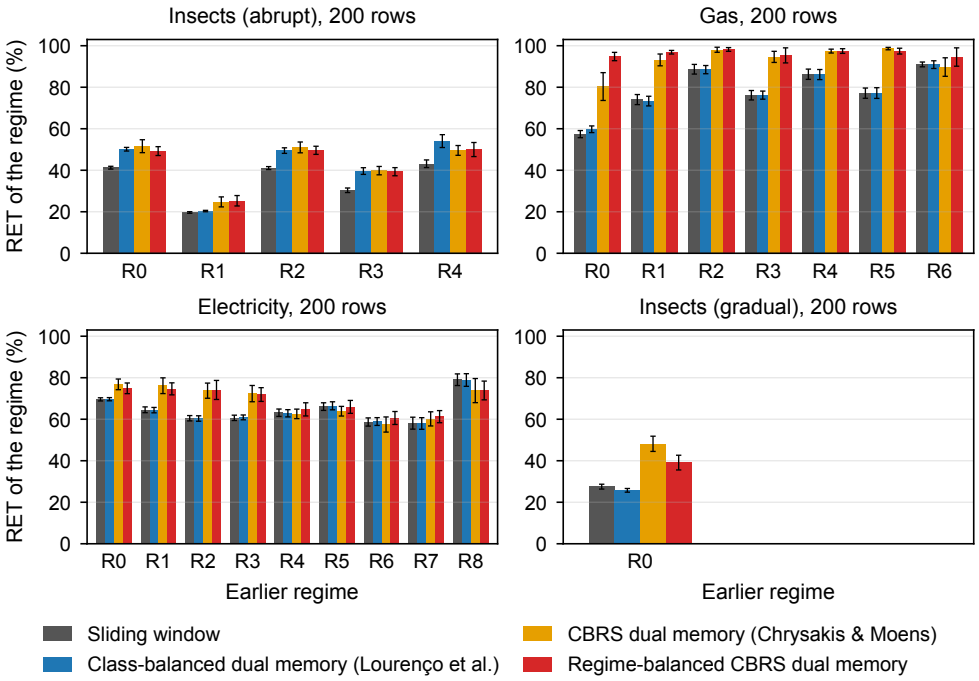



**Figure 8: Retention of each earlier regime at 200 context rows (official RET per regime; mean $\pm$ SD, ten seeds) for the sliding window, the class-balanced dual memory (Lourenço et al. [35]), the CBRS dual memory (after Chrysakis and Moens [5]) and the regime-balanced CBRS dual memory. On Insects (gradual) only R0 has ended. Regimes near the end of a stream are measured only briefly after they end.**

the CBRS dual memory, the regime-balanced CBRS dual memory retains more on Gas ($+1.84$ and $+0.87$ pp, 10/10 and 9/10), as much on Electricity ($+0.37$ and $-0.30$ pp), where it adapts better ($+1.06$ and $+0.54$ pp in ADAPT, 8/10 and 10/10), and less on Insects (abrupt) ($-1.24$ and $-0.96$ pp) and Insects (gradual) ($-1.74$ and $-0.23$ pp); the differences on Gas, in ADAPT on Electricity, on Insects (abrupt) at 1000 rows and on Insects (gradual) at 200 rows are significant after Holm correction. Figure 8 shows where. On Gas, whose regimes range from 278 to 3,313 rows, the CBRS dual memory keeps few rows of the short regimes and loses accuracy on some of them (R0, 344 rows: 80.3 against 94.8 % at 200 rows); at 200 rows it holds no row of the short Insects (abrupt) R4 in 53 % of the updates after it ends (Gas R3: 29 %), although it retains about as much there as the regime-balanced CBRS dual memory, which keeps rows of every regime by construction. On Insects (gradual) the CBRS dual memory retains more of the old regime (48.2 against 39.1 % at 200 rows, 55.0 against 50.4 % at 1000, in every seed and significant after Holm correction), as written down before round 6: it gives a full class share to two classes frequent in R0 but rare in R1. In the worst-retained regime ($\text{RET}_{\text{worst}}$, post hoc), the regime-balanced CBRS dual memory retains at least as much as the CBRS dual memory in all six combinations with more than one ended regime, significantly on Gas at both budgets and on Insects (abrupt) at 1000 rows (Appendix D).

*Cost in adaptation.* Keeping old rows costs adaptation where the target depends on the recent past (Figure 7c). Against the class-balanced dual memory, ADAPT falls most on Electricity at 200 rows, by 4.45 pp with the CBRS dual memory, 3.39 pp with the regime-balanced CBRS dual memory and 2.10 pp

with the regime-balanced dual memory, and by 0.95 to 1.33 pp on Insects (gradual) with the two CBRS memories; on Gas and Insects (abrupt) it changes by at most 0.25 pp.

*Without given boundaries.* When the regime-balanced dual memory closes regimes at ADWIN alarms instead of the documented change points, its gain depends on how well the detector finds the changes (Table 9 and Figure 10 in Appendix D). On Gas it keeps most of its gain over the class-balanced dual memory, $+7.58$ and $+9.02$ pp in $\text{RET}_{\text{seg}}$ (10/10), and on Insects (gradual) $+2.24$ and $+2.83$ pp. On Electricity it loses the gain at 200 rows ($-2.34$ pp) and adapts 5.47 pp worse: the seasonal changes do not stand out in the model's error rate, so the detector cannot find them, and its many alarms split the long-term part into short pieces. This is a problem of detection, not of allocation. On Insects (abrupt) the variant retains slightly less than the class-balanced dual memory ($-1.21$ and $-0.61$ pp).

*Registered hypotheses.* Of the eight hypotheses of round 2 (the 1000-row regime-balanced dual memory against the 200-row baseline and other policies on Insects (abrupt)), five hold and three fail (Table 8 in Appendix C): H3, because the RET gain over the class-balanced dual memory ($+1.42$ pp) is below the required 2 pp; H4, because after the second and fourth change points the regime-balanced dual memory recovered more slowly than the 200-row baseline, not faster (Section 5.3); and H6, because the gain over the 66 most recent rows of each class ($+0.81$ pp) is below 2 pp. Among those that hold, no window size from 100 to 2000 rows matches the regime-balanced dual memory in both ADAPT and RET (H5, Figure 14 in Appendix D). Varying the short-term share of the 200-row memories (round 4, exploratory) left the sign of the official-RET gain of the regime-balanced dual memory over the class-balanced dual memory unchanged in all twelve combinations and every seed, but not its size (on Gas $+18.21$, $+18.00$ and $+14.14$ pp at shares 0.65, 0.75 and 0.85; Figure 16 in Appendix D); in $\text{RET}_{\text{seg}}$ it stays positive in every seed on Gas, Electricity and Insects (gradual) ($+1.07$ to $+13.45$ pp) but lies between $-0.20$ and $+0.51$ pp on Insects (abrupt).

*Answer to RQ3.* Keeping old rows is the main factor: each of the three memories that keep them retains much more than the class-balanced dual memory where that memory collapses or quickly replaces old regimes. How the long-term part is shared among regimes is secondary, and no memory retained most on every stream: equal shares helped where the regimes differ much in length (Gas), while a class-wise reservoir helped where class proportions shift between regimes (Insects (gradual)) or the regimes are long (Insects (abrupt) at 1000 rows); which to choose for which stream is a hypothesis to be tested on further streams. Keeping old rows costs adaptation where the target depends on the recent past, most on Electricity.

## 5 DISCUSSION

### 5.1 Why Keeping Old Rows Reduces Forgetting

On Gas, a memory that keeps old rows of the earlier regimes needs only a few of each: at 200 rows, six to twelve rows per regime keep the accuracy on earlier regimes at about 95–98 % in the regime-balanced CBRS dual memory (Figure 8) and in the first version, against 57–91 % for the sliding window and the class-balanced dual memory. A linear classifier separates each Gas regime from the later ones almost perfectly (cross-validated AUC 0.992–1.000, where 0.5 means inseparable), plausibly because sensor ageing moves the features into regions that later data no longer cover, so a handful of a regime's own rows suffices. The gap on Gas is therefore mostly that between keeping some old rows and keeping none, not a matter of how they are shared: the CBRS dual memory, which ignores the regimes, already retains

11.2 and 11.0 pp more than the class-balanced dual memory in $\text{RET}_{\text{seg}}$, against 13.4 and 11.7 pp for the regime-balanced dual memory.

### 5.2 When Do Equal Shares Help, and What Do Old Rows Cost?

Whether equal shares per regime or a reservoir retained more appeared, on our streams, to depend on regime lengths and class changes: equal shares protect short regimes, which a small reservoir can leave without rows, and a reservoir keeps more of long regimes and of classes that become rare later (Insects (gradual)). The explanations for short regimes and for Insects (gradual) came after round 5 and were written down before round 6, the one for long regimes only after round 6; a rule for choosing would need streams chosen for the purpose.

The cost in adaptation depends on whether old rows help or mislead on the current regime. It is largest on Electricity, whose label states whether the price is above its mean over the previous 24 hours, a quantity not among the features; older rows reflect other price levels, and even the 1000-row window adapts 4.90 pp worse than the 200-row one. At 200 rows the median age of the long-term rows is about 3.7 days in the class-balanced dual memory but about 176 days in the regime-balanced dual memory, which costs 2.10 pp; the two CBRS memories, whose long-term rows come from the whole past, cost more (Section 4.3). Repeating the previous label row by row even scores 84.86 % on Electricity, more than any memory here (at most 82.6 %), because neighbouring labels are strongly correlated (Appendix B). This stream suits sequence models better than a learner that sees labels in batches; we use its adaptation results only to examine the mechanism of the cost, and on it retention trades against adaptation.

### 5.3 When Is Retention Useful?

Our streams contain no controlled recurrence, so accuracy on the stream could not show the benefit of retention, and it did not. Once its labels are revealed, the frozen model learns a new regime from its rows, even one that resembles an earlier one, a small recent context fastest: the pre-registered expectation that kept regimes speed up recovery (H4) failed, mainly as an effect of context size. On synthetic streams with recurring concepts, Wu et al. [57] found that a context built from a concept's past rows predicted it almost perfectly right after its return, where a FIFO context did not; this is the moment that RET stands in for. Retention therefore pays off when an earlier regime must be predicted before its new labels arrive, and costs when the target depends only on recent history.

### 5.4 Sustainability and Ethics

The choice of memory also matters for the three dimensions of sustainable development. Economically, a frozen model needs no retraining when the data change, so adaptation costs only prediction, which grows with the budget; a small budget can suffice (on Gas the regime-balanced CBRS dual memory retained 97.39 % with 200 rows and 98.93 % with 1000, in $\text{RET}_{\text{seg}}$). Ecologically, the GPU experiments ran for about 242 hours summed over runs (energy use was not measured) and neither pre-trained nor fine-tuned a foundation model; the smallest budget that gives the required retention is preferable. Socially, a memory that keeps earlier regimes is more dependable when conditions return, as in energy and environmental monitoring, and as a set of rows it is easier to audit than knowledge stored in weights.

Forgetting and keeping both raise ethical questions. A memory that keeps only recent rows may serve rare or returning conditions worse, unseen in the accuracy on the stream; if regimes corresponded to groups of people, the forgetting would affect them unequally, and equal shares per group would be one

way to protect a small group. Our streams contain no such groups, so this is a possibility, not a finding. Keeping old rows is not harmless either: they can encode relations that no longer hold, as on Electricity, and where rows describe people, keeping them longer conflicts with the principle that personal data be kept no longer than necessary (EU General Data Protection Regulation, Regulation (EU) 2016/679, Art. 5(1)(e)). Which part of the past a model keeps should be a documented decision, not a side effect of the memory policy.

### 5.5 Relation to Existing Work

Verma et al. [53] found keeping only the most recent rows as effective as the dual memory in prequential accuracy; recency collapse explains this where the classes arrive mixed, since the class-balanced store then predicts, and as retention shows also forgets, like a window. Its long-term store aims to keep recurring classes, not regimes [35], so the collapse is the measured form of its authors' remark that the design addresses forgetting of classes only [34]; replacing its oldest-first eviction by class-balanced reservoir sampling [5], a single rule, raises $\text{RET}_{\text{seg}}$ by 3.4 to 11.2 pp on three of our streams. In replay for continual learning, a reservoir is a strong default and balancing by group pays off mainly with few slots per group [4]; here, balancing the reservoir by regime raised $\text{RET}_{\text{seg}}$ significantly only on Gas. Wu et al. [57] keep separate contexts per concept, each with its newest rows; our second level of forgetting suggests sampling such stores over the whole concept. Finally, the cost on Electricity mirrors the freshness effect of Wang et al. [54], where the most recent labels helped a corrector of a frozen tabular model and the oldest harmed it; retention shows the other side, as rows stale for the current regime are the only evidence the model keeps about earlier ones.

### 5.6 Limitations

- *Method and metric.* The two regime-balanced memories use the documented regime boundaries, and the official RET averages over exactly these regimes. $\text{RET}_{\text{seg}}$ removes the equal weight per regime, but the holdout rows are still drawn per regime, so a short regime weighs much in its segment. The boundary-free variant exists only for the first regime-balanced memory, and its gain depends on how well ADWIN finds the changes (Section 4.3). On Electricity and Gas the regime-balanced memories were given only the documented cut (seasons, recording batches); a finer or coarser cut, such as months, was not tried, so the effect of the regime grain is untested.
- *Exploratory analyses.* The CBRS dual memory, $\text{RET}_{\text{seg}}$, the boundary-free runs on Gas, on Electricity and on Insects (abrupt) at 200 rows, and the regime-index control were added after a review of the first draft (Section 3.1), and the regime-balanced CBRS dual memory was designed after those results and run on the same four streams, so it may suit them better than others. All are exploratory; $\text{RET}_{\text{worst}}$ and the analyses of memory content were made after the results (Appendix C).
- *Evidence base.* Four streams, one per kind of drift, so the kind of drift is confounded with the label definition, class balance, regime lengths and separability; one backbone; two budgets. The seeds vary only the holdout sets, TabPFN's random state and the random draws of the two CBRS memories, and each stream has one order; the round-3 runs at 1000 rows on Insects (abrupt) are those of round 2, not an independent replication; on Insects (gradual), the rows kept for R0 partly come from the transition (Appendix B).

- *Baselines.* The class-balanced dual memory is the rule of Lourenço et al. [35] scaled to our budgets (their grid was 600 to 1000 rows with TabPFN version 1; batches and metric differ). The regime-index control can only show that the index alone does not replace old rows (Appendix D).
- *Interpretation.* RET is a proxy for recurrence; recovery after a true return was not measured, and the explanations of the per-regime results are hypotheses from four streams.

## 6 CONCLUSIONS AND FUTURE WORK

With frozen weights, the context of a tabular foundation model is its only memory of a stream, and a bounded context must decide what to forget. Scoring retention next to adaptation for a frozen TabPFN on four public streams, we reach a main conclusion in three parts, one per research question (Table 6). First, forgetting is large, and accuracy on the stream does not show it. Second, it comes from evicting the oldest rows, at two levels: within a class in the class-balanced dual memory, which on Gas and Electricity keeps almost only the newest rows (recency collapse), and within a regime in our first regime-balanced memory, which then keeps only the end of each regime. Third, what reduces forgetting is keeping old rows at all; how they are shared among regimes is secondary.

*Contributions.* The thesis makes the forgetting of a frozen tabular foundation model visible, explains it by the eviction rule, and shows in a controlled comparison of four long-term rules that keeping old rows at all matters more than how they are shared among regimes, at a cost in adaptation that depends on the stream. The principles come from replay memories in continual learning [4, 5, 31]; what is new is their study for a frozen model's context, with drift regimes as the groups.

*Implications for practice.* On the evidence of one TabPFN backbone, a long-term store should not evict its oldest rows first. A class-balanced reservoir, which needs no regime boundaries, retained well on our streams and is a candidate default; where boundaries are known and the regimes differ much in length, equal shares per regime may pay off. Both are hypotheses that further streams should test. Without known boundaries, the gain of a regime-balanced memory depends on how well a detector finds the changes: with ADWIN the first design kept most of its gain over the class-balanced dual memory on Gas, but not on Electricity, whose seasonal changes do not stand out in the model's errors. Where the target depends mainly on the recent past, keeping old rows trades adaptation for retention.

**Table 6: Answers to the research questions. Retention is $\mathrm{RET}_{\mathrm{seg}}$ unless stated; ranges cover both context sizes.**

| Question | Answer | Evidence |
|---|---|---|
| RQ1: How much is forgotten, and does stream accuracy show it? | Much, and stream accuracy hides it: the four memories differ by at most 4.5 pp in ADAPT but by up to 13.3 pp in $\mathrm{RET}_{\mathrm{seg}}$. With a 200-row window the model scores 35.08 % in RET on earlier regimes of Insects (abrupt), about twice chance, against 75.68 % on the stream. The forgetting appears in RET and in $\mathrm{RET}_{\mathrm{seg}}$. | Table 5, Figure 5 |
| RQ2: Why do memories forget? | They evict the oldest row of a group. The class-balanced dual memory does so within a class; where the classes arrive mixed (Gas, Electricity), it keeps almost only the newest rows. The first regime-balanced memory does so within a regime and keeps only the end of each regime. | Figures 1 and 6 |
| RQ3: What matters more for retention, and at what cost? | Keeping old rows: the three memories that do so retain 2.2 to 13.4 pp more than the class-balanced dual memory on Gas, Electricity and Insects (gradual), in every seed, and about as much on Insects (abrupt). Sharing among regimes is secondary, and no memory retained most on every stream. Keeping old rows costs adaptation where the target depends on the recent past: on Electricity at 200 rows, 3.3 to 3.8 pp more retention against 2.1 to 4.5 pp less adaptation. | Figures 7 and 8 |

*Future work.* In order of importance: (1) streams in which earlier situations really return, to measure whether the kept rows make the model recover faster (RET is only a proxy for this); (2) streams chosen to test when equal shares per regime help and when a class-wise reservoir does, and rules that combine both, such as sharing first by class and then by regime; (3) detecting regime changes from signals better suited to a frozen model than its error rate, or taking them from known events; (4) other backbones (TabICL [47], the original TabPFN [20] used by Lourenço et al. [35]), more streams per kind of drift, and regression; (5) a short/long split that adapts to the stream, to reduce the cost on Electricity.

*Take-home message.* For a frozen tabular foundation model, the rows in its context are all it remembers of the stream. On the evidence from TabPFN, when the past may return, a long-term store should keep a sample of the whole past rather than evict its oldest rows first, and retention should be reported next to stream accuracy.

## ACKNOWLEDGMENTS

I am deeply grateful to my supervisor, Tianyu Wang, for his careful guidance throughout this degree project: for pointing out a promising direction and for his expert advice at every stage. I also thank KTH Royal Institute of Technology for my years of study here. The teachers of the courses I took were of a very high standard, and what I learned from them gave me the solid foundation on which this project was built.

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## A PRELIMINARY STUDY: THREE-LEVEL SYSTEM AND DRIFT DETECTION

This appendix summarises the preliminary study that preceded the change of direction (Section 3.1).

*The three-level system.* A frozen TabPFN with a 200-row sliding context gives a slow prediction. A fast corrector (nearest neighbours or an exponential average over the residuals of recently labelled rows) and a small gated adapter, one per regime in later versions, add corrections; a gate weights the three outputs. A drift detector (ADWIN on the model's errors) can trigger an action on an alarm: switching to or creating an adapter, resetting the context to its most recent rows, or clearing the corrector's buffer.

*Synthetic streams.* Three generators were used: *rotating boundary* (two Gaussian features and a linear boundary that rotates slowly; gradual drift), *regime switching* (three regimes with independent feature means and decision weights, cycling every 500 rows; abrupt drift) and *combined drift* (five features shift slowly while the boundary changes abruptly every 3,000 rows).

*Results.* Table 7 lists the main results. Paired comparisons on synthetic streams use five seeds, each of which generates a different stream; on the real streams the five runs per segment used the same data and differed only in the initialisation of the adapter, so their $p$-values do not describe independent replications.

Two observations of round 1 shaped the later design. Each change point of the Insects segments came at the end of a run of a single class (34 to 538 rows), and the alarms mostly followed these changes of class proportions rather than the documented change points. And even a perfect action had little room to help: after two of the three change points, the excess errors over the new regime's steady state were below what the test could detect. The event-triggered system thus gave no measurable gain and could not be tested well on these segments, whereas retention of earlier regimes does not depend on the short period after a change. This motivated the study of the context itself.

## B DATASET DETAILS AND REPRODUCIBILITY

This appendix gives the source, preprocessing, regimes and holdout sets of each stream (Table 3) and the information needed to reproduce the results. The three streams added in round 3 met the criteria fixed in the pre-registration: regimes definable from documented or natural time boundaries, enough rows of most classes in every regime for a holdout set, later regimes that differ from earlier ones, a size within TabPFN's limits, and no columns that leak the label. Content hashes of these streams are in the pre-registration in the public repository.

**Table 7: Main results of the preliminary study. Differences are in plain (not balanced) accuracy, in percentage points, relative to TabPFN with a sliding context alone; "sig." marks a paired $t$-test beyond the critical value ($n = 5$), "n.s." one within it.**

| Experiment | Result |
|---|---|
| Fast corrector alone (synthetic) | $+0.04$, $+0.12$, $-0.14$ on regime switching, rotating boundary and combined drift; none significant. |
| Full three-level system (synthetic) | $+1.00$ (sig.) on rotating boundary, $-0.18$ (n.s.) on regime switching, $-0.47$ (sig.) on combined drift. |
| Context reset at the true change points (synthetic, regime switching) | $+0.51$ (sig.; better in 5/5 seeds), $+2.36$ on the rows after the changes. |
| Five variants of detector input and adapter (synthetic) | combined drift $-0.28$ to $-0.55$ in all five; rotating boundary $+1.32$ to $+1.52$ (sig.). In the three variants with alarms, all 52 routing decisions created a new adapter; an existing one was never reused. |
| Three-level system on real streams (CPU) | Electricity $-0.06$ (n.s.); Insects $-0.17$ (sig.), on segments that later turned out to cover only two of the five documented change points. The own detector raised no alarm in 20 Insects runs and one in 15 Electricity runs. |
| Detector replay on the saved error streams | The own ADWIN used a bound without the empirical variance and needed a shift of the error rate of at least 0.209 for a 200/200 split. The river ADWIN ($\delta = 0.002$) on the same errors raised alarms in 16 of 20 Insects runs, detected every covered change point (median delay 266 rows) and stayed nearly silent on Electricity (1 of 15 runs). |
| Round 1: GPU, Insects, river ADWIN, one seed | Actions after an alarm changed the errors on a segment by $-6$ to $+5$ (smallest $p = 0.36$; McNemar), also when triggered at the true change points. With no action, the three-level system made 26 and 23 more errors than TabPFN alone on two segments ($p = 0.005$ and $0.001$, sign test) and 4 fewer on the third ($p = 0.72$). |

*Insects (abrupt).* The balanced abrupt stream of the INSECTS benchmark [50]: 52,848 rows, 33 features, 6 classes. The paper states no licence for the data; we use them as a public research benchmark. Its documented change points, rows 14,352, 19,500, 33,240, 38,682 and 39,510, define six regimes, R0 to R5 (changes of temperature). Each seed draws 50 rows per class from each regime (20 per class from the short R4), uniformly over the whole regime, as holdout set (1,620 rows); in the remaining stream of 51,228 rows the change points lie at 14,052, 18,900, 32,340, 37,482 and 38,190. In the full data set, before the holdout sets are removed, each change point is preceded by a run of a single class (1,754, 74, 97, 537 and 34 rows), and the data end with one of 822 rows; in the stream these runs are slightly shorter. Adaptation is therefore also reported with all runs of at least 30 identical labels removed (*masked* ADAPT, used in hypothesis H2); the run before the fifth change point reaches 30 rows in only four seeds. The masked values are in the public repository. In the boundary-free variant (Section 3.3, 1000 rows), ADWIN raised 13 to 15 alarms per seed, found the three earliest change points within 11 to 79 rows, and raised its other alarms elsewhere.

*Gas.* The Gas Sensor Array Drift data of Vergara et al. [52] (UCI dataset 224, also OpenML 1476; licence CC BY 4.0): 13,910 rows of 128 sensor features and 6 gases, recorded in ten batches over 36 months. The source file begins with the last batch, so the batches are put back in time order; each covers one to five recording months. Batches with fewer than 300 rows are merged with the next (batches 4 and 5, 8 and 9), giving eight regimes. The order of rows inside a batch is grouped by label in the source and undocumented, so it is shuffled once with a fixed seed; every run uses the same order. Holdout sets here and in the two following streams take, per regime and class, the smaller of 50 rows and a quarter of the eligible rows, and no rows when this would be fewer than 10; this gives 1,635 rows on Gas.

*Electricity.* The New South Wales electricity market data described by Harries [18] (OpenML dataset 151, listed there as public), 943 days of 48 half-hours (45,264 rows) after removing day 363, a near-duplicate of day 362. The features are the day of the week, the half-hour, the price and the demand in New

South Wales. The date column is dropped because, with labels revealed in batches, it acts as a time index into strongly autocorrelated labels; the three columns for Victoria and the transfer between the states are dropped because they are constant placeholders for the initial 17,424 rows. The label states whether the price is above its mean over the previous 24 hours [18]. The regimes are the meteorological seasons of the southern hemisphere, with partial seasons shorter than 30 days merged into their neighbour: ten regimes, 1,000 holdout rows.

*Insects (gradual).* A slice of the imbalanced incremental-gradual INSECTS stream around its documented change point: 10,000 rows at 35 °C, a transition of 3,000 rows and 10,000 rows at 23 °C. Matching every row to the temperature-ordered version of the data showed that in the transition instances of both temperatures are interleaved, with about 30, 50 and 70 % at 23 °C in its three thirds. R0 is the 35 °C part with the transition, R1 the 23 °C part; transition rows are not used for holdout sets, and R0 has none for one class with only ten pure rows (550 holdout rows). Since the end of R0 is ambiguous, the memory was closed in three ways: (a) at the documented change point, the main arm used in Sections 4 and 5; (b) at the start of the transition, which is known only from the analysis above and so cannot be used in deployment; and (c) by ADWIN alarms. Evaluation regimes and holdout sets are the same in all arms (Figure 9).

All three arms retain more than the class-balanced dual memory, but by different amounts. At 200 rows, the RET of the regime-balanced dual memory is 36.89 % in arm (a), 33.17 % in arm (b) and 34.89 % in arm (c), against 25.74 % for the class-balanced dual memory. At 1000 rows it is 46.99, 50.78 and 43.04 %, against 31.36 %. Closing R0 at the start of the transition is thus not an upper bound: at 200 rows, arm (b) retains less than arm (a). The arms also keep different rows for R0: at the end of the stream, arm (a) holds rows from the end of the transition and arm (b) rows of the pure 35 °C part. In the main arm, the rows that the regime-balanced memories keep for R0 thus come partly (in the first version entirely) from the transition, not from the pure 35 °C part on which R0 is tested. Arm (c), with ADWIN, is the least stable. At 200 rows its RET has an SD of 5.25 pp over the seeds, and at 1000 rows its ADAPT is 0.82 pp below that of the class-balanced dual memory, in every seed.

*Trivial baselines.* The two baselines of Section 4 use only the revealed labels: the last label revealed before a batch, and the majority class of the last 200 revealed rows. The lowest ADAPT of the memories run on all four streams (those of Table 5, the regime-balanced dual memory, its boundary-free variant and the regime-index control) lies far above the better of the two on every stream: 74.67 against 31.36 % on Insects (abrupt), 98.54 against 21.13 % on Gas, 76.54 against 54.01 % on Electricity and 68.80 against 28.78 % on Insects (gradual). The pre-registration defined them row by row, which uses labels of the current batch that the model has not seen; on Electricity, repeating the previous row's label then scores 84.86 % balanced accuracy, because neighbouring labels are strongly correlated. With labels revealed per batch, as for the model, it scores 54.01 %.

### Reproducibility

All main runs used Python 3.12 with the package versions fixed in `requirements-lock.txt` (among them tabpfn 6.4.1 with the TabPFN-2.5 weights, torch 2.10.0 with CUDA 12.8, river 0.23.0 and scikit-learn 1.3.2) on one laptop with an NVIDIA RTX 2060 Max-Q GPU (6 GB). The public repository (https://github.com/wenboxia/tabular-fm-temporal-shifts) contains the code, the unit tests,

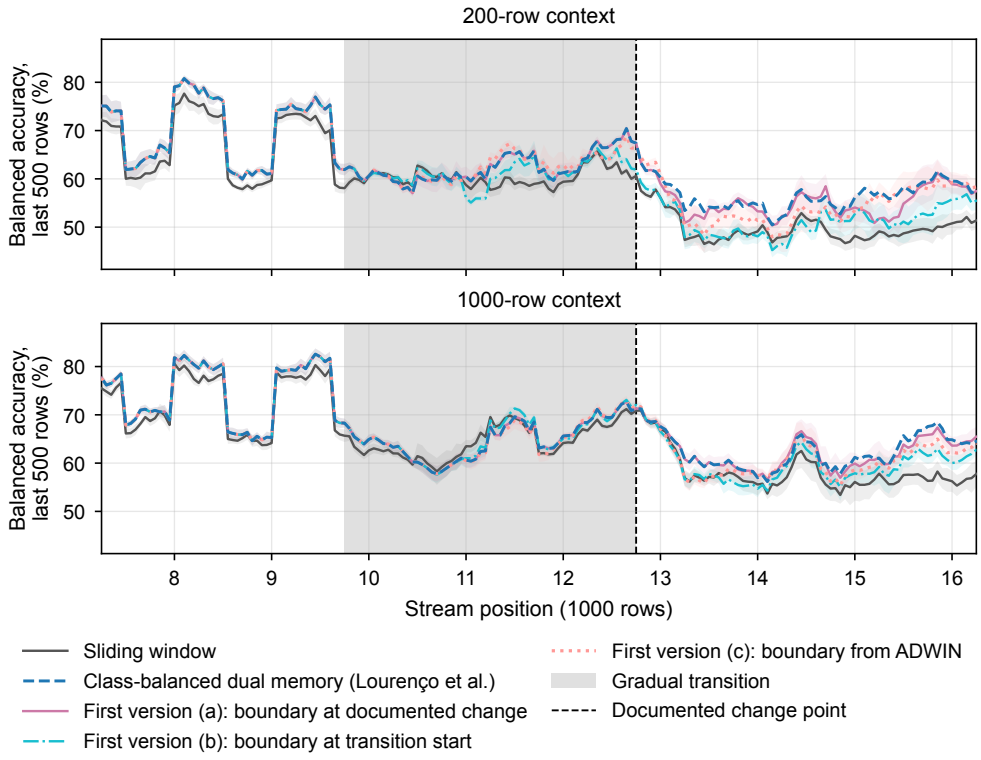



**Figure 9: Accuracy around the gradual transition of Insects (gradual), measured over the last 500 rows (mean ± SD, ten seeds; top: 200 rows, bottom: 1000 rows). Grey: the 3,000-row transition in which both temperatures are mixed; dashed: the documented change point. Arms (a)–(c) of the regime-balanced dual memory close R0 at the documented change point, at the start of the transition and at ADWIN alarms. Before the change point, the class-balanced dual memory and arm (a) hold the same rows; afterwards, the sliding window is mostly the lowest curve. Because each curve looks back 500 rows, a change shows up to 500 rows after it happens.**

the pre-registration with its amendments, the result reports of rounds 2 to 4 with their supplementary tables and figures, and the per-seed results (`results/round{2,3,4,5}_per_seed.csv`; the last file holds every run of rounds 5 and 6 and the round-2 and round-3 runs used in the same comparisons, as computed by `scripts/analyze_round5.py` from the raw runs). The scripts `scripts/reproduce_round{1,2,3,4}.sh` rerun rounds 1 to 4; with `SMOKE=1`, those of rounds 1 to 3 run a short check on a CPU in a few minutes. Rounds 2 to 6 use seeds 0 to 9 and round 1 seed 42. The raw result files are too large for the repository and are kept by the author.

*Checks.* No context contained a future row or exceeded its budget. Replaying all 360 dual-memory runs used in rounds 3 and 4 (20 of them reused from round 2) and all 320 dual-memory runs compared in rounds 5 and 6 from the labels alone reproduced the memory fingerprint at every checkpoint, and the GPU runs reproduced the CPU pilot on 99.3–99.6 % of its predictions. With batches of 10 instead of 50 rows (Insects (abrupt), three seeds), the RET gain of the 1000-row regime-balanced dual memory over the 200-row baseline was nearly unchanged (+12.29 against +12.21 pp).

## C PRE-REGISTRATION AND HYPOTHESES

The pre-registration and its amendments are in the public repository (`results/phase56_prereg.md`); each version was committed before the runs it governs, and later changes were added as dated amendments with their reasons.

*Original registration (before round 2).* It fixed the question, the Insects (abrupt) stream, the holdout sets, the protocol of Section 3.2, the metrics, fourteen policies, eight hypotheses and a go/no-go criterion for a pilot run on a CPU. The proposed method was a *regime archive*: a class-balanced sample of 25 rows per class from each finished regime, added to a 200-row window only when a query resembled that regime more than the window.

*Amendment 1 (after the pilot).* The archive method failed the go criterion: its ADAPT was 2.71 pp below the 200-row window (limit $-2$ pp), although its RET was 8.93 pp higher. Queries were routed to old archives for 13–72 % of the current regime's rows, and predictions on those rows were worse than with the window alone. The pilot also showed the long-term part of the class-balanced dual memory drifting towards the latest regimes. The amendment therefore added the regime-balanced dual memory and its boundary-free variant (16 policies in all) and made the former the proposed method, with the hypotheses unchanged except that H7 became a non-inferiority test against the class-balanced dual memory, with a margin of $-0.5$ pp in ADAPT. The new method passed the same go criterion ($+12.6$ pp RET, $+0.93$ pp ADAPT for the 1000-row memory against the 200-row baseline) before any main run.

*Amendment 2 (before the main runs).* It placed H2 in the Holm family with H1 and H3, required all ten seeds for a formal verdict, and stated the ADWIN settings and the reported quantities.

*Amendment 3 (round 3).* Written before any full run of round 3, it added the three further streams with the selection criteria listed in Appendix B, the holdout rule, the three arms of the gradual stream, the comparison of the three policies at 200 and 1000 rows, and the analysis: a Holm family of the eight RET comparisons between the regime-balanced dual memory and the class-balanced dual memory, the $-0.5$ pp margin for ADAPT, and no success thresholds. Pipeline checks and a few CPU runs made before the GPU runs, among them retention on Gas for some policies, are listed in a note to the amendment; nothing registered was changed because of them. The separability measure of Section 5.1 was defined after those Gas runs and is exploratory.

*Note on the short-term share (round 4).* Written after the results of round 3 and before any run of round 4, it registered the shares 0.65 and 0.85 at 200 rows as an exploratory analysis without tests or thresholds.

*Amendments 4 and 5 (after the first draft; exploratory).* After a review of the first draft, Amendment 4 was registered before any run of round 5. It added the boundary-free variant on Gas and Electricity at both budgets and on Insects (abrupt) at 200 rows, the CBRS dual memory, the class-balanced dual memory with the regime index as an extra feature, and $\mathrm{RET}_{\mathrm{seg}}$, with segments of 1,000 and 5,000 rows and a lag of 5,000 rows as sensitivity analyses. $\mathrm{RET}_{\mathrm{seg}}$ needs no new runs and had been computed from the stored predictions before the amendment was written, which the amendment discloses together with those numbers. Four families of comparisons were defined, each covering one pair of memories over the four streams and the two budgets (eight tests), Holm-corrected within the family and separately for each metric, without success thresholds: the boundary-free variant against the class-balanced dual

memory (A), the regime-balanced dual memory against the CBRS dual memory (B), the regime-index control against the class-balanced dual memory (C1) and the regime-balanced dual memory against the control (C2). Readings fixed in advance labelled the outcomes. (A) On Gas, a gain in RET of the boundary-free variant over the class-balanced dual memory of at least 10 pp would largely rule out a dependence on the documented boundaries there; it was reached at 1000 rows ($+11.11$ pp) but not at 200 ($+9.66$ pp). Two caveats apply to this reading: RET still scores the documented regimes, and in every run, the boundary-free ones included, a batch never spans a documented change point. (B) Where the regime-balanced dual memory exceeds the CBRS dual memory by at most 0.5 pp in $\text{RET}_{\text{seg}}$, its gain is attributed to keeping old rows rather than to sharing by regime; this held in six of the eight cases, all but Gas. (C) If the control reaches half of the gain of the regime-balanced dual memory, the boundary information itself would explain much of that gain; it was reached in none of the seven cases where the reading applies, a reading registered as one-sided (Appendix D). Amendment 5 was registered after the results of round 5 and before any run of round 6. It added the regime-balanced CBRS dual memory, registered as the regime-balanced reservoir dual memory and designed from the finding that the regime-balanced dual memory keeps only the newest rows of each regime, with two families of comparisons, against the CBRS dual memory (E1) and against the regime-balanced dual memory (E2), $\text{RET}_{\text{seg}}$ as the main measure, $\text{RET}_{\text{worst}}$ as a further measure, and the expected outcomes written down before the runs; E1 and E2 have the same form as A to C2. The other comparisons of rounds 5 and 6 in the main text, among them the three memories that keep old rows against the class-balanced dual memory, were registered as descriptive, outside the families (except the class-balanced dual memory against the sliding window, added after the results), and are reported with sign counts. All comparisons of rounds 5 and 6 are exploratory and use $\text{RET}_{\text{seg}}$ as their main measure; the official RET, the measure of the comparisons registered for rounds 2 and 3, is reported beside it and decides only reading (A).

*$p$-values of round 3.* The eight pre-registered RET comparisons between the regime-balanced dual memory and the class-balanced dual memory (two-sided paired $t$-tests over the ten seeds, Holm-corrected within the family) gave adjusted $p$-values of $1.05\times10^{-6}$ on Insects (abrupt) at both budgets, $4.87\times10^{-12}$ and $1.21\times10^{-9}$ on Gas, $9.66\times10^{-9}$ and $5.02\times10^{-8}$ on Electricity, and $1.21\times10^{-10}$ and $4.87\times10^{-12}$ on Insects (gradual), at 200 and 1000 rows. They reflect only the two sources of randomness that differ between seeds, the sampling of the holdout sets and the random state of TabPFN; each stream has one fixed order (Section 5.6), and the long-term memories of different seeds hold nearly the same rows (shown for Insects (gradual) in the round-3 report of the public repository). The main text therefore reports the number of seeds with the same sign. Intervals and unadjusted values are in the round-3 report of the public repository.

*Hypotheses of round 2.* Table 8 gives the hypotheses as amended and their outcome. H1 to H3 are primary: one-sided paired $t$-tests over the ten seeds, Holm-corrected, $\alpha = 0.05$, with the sign required in at least 9 of 10 seeds. H2 tests non-inferiority with a margin of $-0.5$ pp: its one-sided test is of the ADAPT difference plus 0.5 pp, taking the larger $p$ of the raw and masked versions, and at least 9 seeds must show a difference above $-0.5$ pp. H1, H2, H4 and H8 compare the 1000-row regime-balanced dual memory with the 200-row baseline; the same-budget comparisons are in Section 4.3. The originally registered archive method, judged by the original H1 to H3, had a RET 10.41 pp above the 200-row window but an ADAPT 3.24 pp below it (H2 fails) and a RET 0.37 pp below the class-balanced dual memory (H3 fails).

**Table 8: Pre-registered hypotheses of round 2 (Insects (abrupt), 1000-row memories, ten seeds) and their outcome. "Ours" is the regime-balanced dual memory. Differences are in percentage points; 10/10 means positive in all ten seeds.**

| | Hypothesis | Compared with | Result | Verdict |
|---|---|---|---|---|
| H1 | Ours retains at least 5 more, significantly. | 200-row window | $+12.19$ RET, 10/10 | holds |
| H2 | Ours adapts no worse: one-sided 95 % lower bound above $-0.5$, also with single-class runs masked. | 200-row window | $+1.16$ ADAPT (bound $+1.08$); masked $+1.75$ ($+1.67$) | holds |
| H3 | Ours retains at least 2 more, significantly. | Class-balanced dual memory | $+1.42$ RET, 10/10, significant | fails |
| H4 | In the 500 rows after a change, ours is at least 5 better after the 2nd and 4th change point and within $\pm 2$ after the 1st and 5th. | 200-row window | $-6.68$, $-8.07$; $-4.21$, $+0.68$ | fails |
| H5 | No window size is at least as good as ours in both ADAPT and RET. | Windows of 100–2000 rows | none | holds |
| H6 | Ours retains at least 2 more. | The 66 most recent rows of each class | $+0.81$ RET | fails |
| H7 | Ours adapts at most 0.5 worse. | Class-balanced dual memory | $+0.19$ ADAPT | holds |
| H8 | The boundary-free variant keeps at least half of the H1 gain and passes H2. | 200-row window | $+11.37$ RET; bounds $+1.11$, $+1.89$ | holds |

## D SUPPLEMENTARY RESULTS

This appendix gives the further analyses of Amendments 4 and 5 (the boundary-free variant on all streams, the regime-index control, the segment length of $\text{RET}_{\text{seg}}$ and the worst-retained regime), the intervals of further paired differences (Table 10) and supplementary figures: the ADWIN alarms (Figure 10), retention over time on Insects (abrupt) (Figure 11), what the long-term part holds over time on Gas and the other streams (Figures 12 and 13), all sixteen policies of round 2 (Figure 14), how well rows of one regime predict another (Figure 15) and the exploratory analysis of the short-term share (Figure 16). Further tables and figures of rounds 2 to 4 are in the result reports of the public repository (`results/phase56_round{2,3,4}.md` and `results/round{3,4}_analysis.md`).

*Boundary-free variant.* Table 9 and Figure 10 give the ADWIN alarms of the boundary-free variant of the regime-balanced dual memory, and Table 9 its difference to the class-balanced dual memory (intervals in Table 10). A check made after the results compares each run with its own alarms shifted cyclically along the stream, which keeps their number and spacing but not their timing. On Insects (abrupt) and Gas the alarms track the changes: an alarm follows within 1,000 rows in 86–88 and 41–49 % of the changes, against 20–24 % after shifting, with a median delay of 28 to 110 rows against about 940 to 2,750. On Gas only 36–39 % of the change points are followed by an alarm before the next one (Table 9; the 41–49 % above are higher because one alarm can lie within 1,000 rows of two close change points), but the detector raises almost no other alarms, and the variant keeps most of the gain; on Insects (abrupt) it adds five to seven alarms far from any change. On Electricity ADWIN raises about 40 alarms for nine season changes, about 24 of them far from any change, so the long-term part is split into about 40 short pieces. An alarm follows every season change before the next one, after a median of about 550 to 700 rows, but so do the shifted alarms (after about 520 to 560 rows): the alarms do not track the season changes. The detector settings were not tuned (computed by `scripts/analyze_round5.py`).

*Regime-index control.* To ask how much the boundary information alone contributes, the class-balanced dual memory was also run with the index of each row's regime as an extra feature (round 5). Relative to the class-balanced dual memory without it, $\text{RET}_{\text{seg}}$ changed by $+0.03$ and $-0.11$ pp on Gas and $0.00$

**Table 9: The boundary-free variant of the regime-balanced dual memory, which closes a regime at each ADWIN alarm. Change pts: documented change points; Alarms: mean (range) over ten seeds; Detected: share of change points followed by an alarm, whatever its cause, before the next change point or the end of the stream; Delay: median over seeds of each run's median number of rows from a detected change point to its first alarm; Far: alarms more than 1,000 rows from any change point (mean); last two columns: paired difference to the class-balanced dual memory in $\mathrm{RET}_{\mathrm{seg}}$ (positive seeds) and in ADAPT (pp). On Insects (gradual) the first alarm falls inside the transition, about 2,400–2,700 rows before the documented change point (counted under Far), and Detected and Delay refer to the second alarm, inside R1.**

| Stream | Size | Change pts | Alarms | Detected (%) | Delay | Far | $\Delta\mathrm{RET}_{\mathrm{seg}}$ | $\Delta$ADAPT |
|---|---|---|---|---|---|---|---|---|
| Insects (abrupt) | 200 | 5 | 10.9 (10–12) | 88 | 110 | 5.1 | −1.21 (1/10) | 0.00 |
| | 1000 | 5 | 14.2 (13–15) | 92 | 52 | 6.9 | −0.61 (1/10) | +0.23 |
| Gas | 200 | 7 | 3.0 (2–5) | 36 | 28 | 0.2 | +7.58 (10/10) | +0.03 |
| | 1000 | 7 | 3.0 (2–4) | 39 | 34 | 0.5 | +9.02 (10/10) | +0.02 |
| Electricity | 200 | 9 | 42.3 (37–46) | 100 | 554 | 24.0 | −2.34 (0/10) | −5.47 |
| | 1000 | 9 | 39.8 (36–43) | 100 | 698 | 23.9 | +0.69 (7/10) | −1.14 |
| Insects (gradual) | 200 | 1 | 2.0 (2–2) | 100 | 1107 | 1.5 | +2.24 (9/10) | −0.04 |
| | 1000 | 1 | 2.0 (2–2) | 100 | 2466 | 2.0 | +2.83 (10/10) | −0.82 |

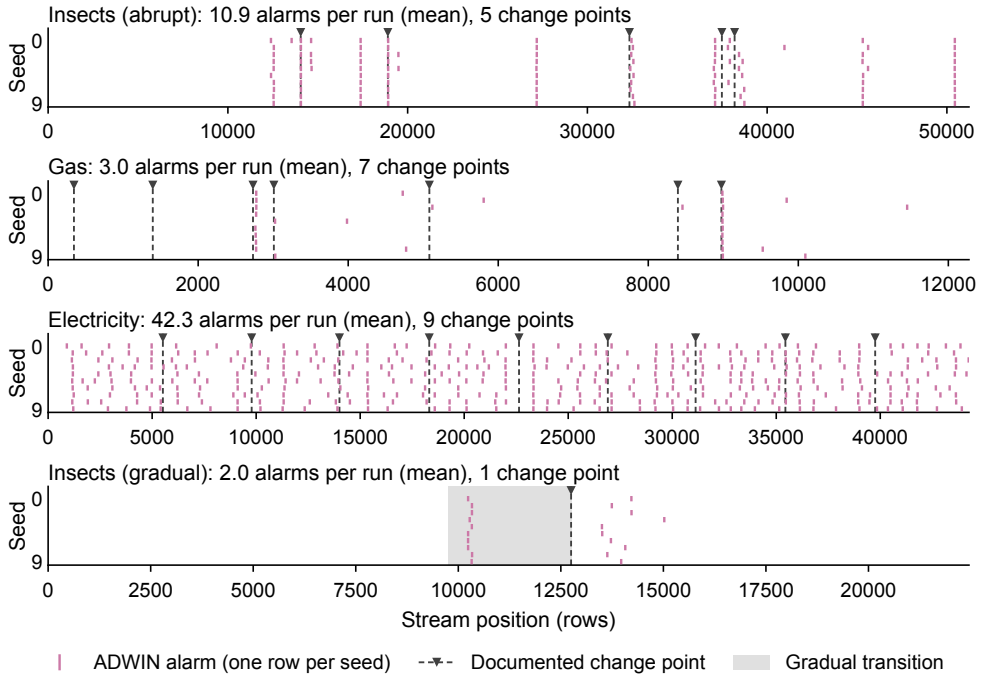



**Figure 10: ADWIN alarms of the boundary-free variant of the regime-balanced dual memory at 200 context rows: one row of ticks per seed (seeds 0 to 9 from top to bottom) at the positions where the memory closed a regime; dashed lines with triangles: documented change points; grey band: the 3,000-row transition of Insects (gradual). On Insects (abrupt) the alarms recur at nearly the same positions in every seed, many just after a change point; on Gas every seed has an alarm shortly after the third or fourth and after the last change point, and few others; on Electricity about 42 alarms per run spread over the whole stream (Table 9).**

and +0.01 pp on Electricity, fell slightly on Insects (gradual) (−0.18 and −0.42 pp, 0/10) and fell on Insects (abrupt) (−2.21 and −3.12 pp, 0/10; intervals in Table 10), while ADAPT rose slightly there

**Table 10: Further paired differences reported in the main text or in this appendix, besides those of Figure 7: mean over ten seeds with the number of positive seeds, and below it the two-sided 95 % $t$ interval (not adjusted for multiple tests), in pp of $\mathrm{RET}_{\mathrm{seg}}$ unless ADAPT is stated. First version: regime-balanced dual memory; Boundary-free: its boundary-free variant (Table 9); Regime index: the class-balanced dual memory with the regime index as an extra feature.**

| Difference | Size | Insects (abrupt) | Gas | Electricity | Insects (gradual) |
|---|---|---|---|---|---|
| First version − CBRS | 200 | −0.30 (4/10) | +2.17 (9/10) | −0.06 (4/10) | −2.74 (0/10) |
| | | [−1.63, +1.03] | [+1.35, +2.99] | [−1.32, +1.20] | [−3.70, −1.78] |
| | 1000 | −1.02 (0/10) | +0.67 (7/10) | −2.47 (0/10) | −1.37 (0/10) |
| | | [−1.42, −0.62] | [+0.12, +1.23] | [−3.08, −1.86] | [−1.91, −0.82] |
| Class-bal. − window | 200 | +9.10 (10/10) | +0.25 (10/10) | 0.00 (5/10) | +2.57 (10/10) |
| | | [+8.85, +9.35] | [+0.10, +0.41] | [−0.10, +0.10] | [+2.28, +2.85] |
| | 1000 | +6.62 (10/10) | +0.52 (10/10) | −0.26 (0/10) | +2.79 (10/10) |
| | | [+6.44, +6.80] | [+0.29, +0.76] | [−0.34, −0.17] | [+2.30, +3.27] |
| Boundary-free − class-bal. | 200 | −1.21 (1/10) | +7.58 (10/10) | −2.34 (0/10) | +2.24 (9/10) |
| | | [−1.82, −0.60] | [+5.82, +9.34] | [−3.38, −1.31] | [+1.33, +3.15] |
| | 1000 | −0.61 (1/10) | +9.02 (10/10) | +0.69 (7/10) | +2.83 (10/10) |
| | | [−0.92, −0.30] | [+7.40, +10.64] | [−0.04, +1.41] | [+2.26, +3.39] |
| Boundary-free − class-bal. (ADAPT) | 200 | 0.00 (6/10) | +0.03 (7/10) | −5.47 (0/10) | −0.04 (6/10) |
| | | [−0.17, +0.17] | [−0.04, +0.11] | [−6.56, −4.38] | [−0.58, +0.50] |
| | 1000 | +0.23 (10/10) | +0.02 (8/10) | −1.14 (0/10) | −0.82 (0/10) |
| | | [+0.17, +0.30] | [−0.01, +0.05] | [−1.34, −0.95] | [−1.10, −0.54] |
| Regime index − class-bal. | 200 | −2.21 (0/10) | +0.03 (7/10) | 0.00 (4/10) | −0.18 (0/10) |
| | | [−2.38, −2.03] | [−0.09, +0.15] | [−0.02, +0.02] | [−0.25, −0.12] |
| | 1000 | −3.12 (0/10) | −0.11 (3/10) | +0.01 (6/10) | −0.42 (0/10) |
| | | [−3.44, −2.80] | [−0.25, +0.04] | [−0.03, +0.05] | [−0.54, −0.31] |
| Reg.-bal. CBRS − CBRS (ADAPT) | 200 | +0.09 (6/10) | −0.03 (4/10) | +1.06 (8/10) | +0.38 (8/10) |
| | | [−0.07, +0.26] | [−0.12, +0.07] | [+0.47, +1.65] | [−0.89, +1.65] |
| | 1000 | +0.16 (9/10) | −0.02 (4/10) | +0.54 (10/10) | +0.13 (6/10) |
| | | [+0.10, +0.22] | [−0.07, +0.02] | [+0.37, +0.72] | [−0.26, +0.51] |

(+0.16 and +0.74 pp). The conclusion is limited to this: giving the model the regime index does not replace keeping old rows. The control is weak, because TabPFN infers each column's type from the context: with a single regime index in the context the column is constant and dropped, with two or three it is categorical and an index absent from the context counts as unknown, and only with four or more is it numerical. In replays from the labels, the column was constant at most checkpoints at which an earlier regime was scored in seven of the eight cases (about 62 of 79 on Insects (abrupt) at 200 rows; on Gas at 1000 rows it was mostly categorical, 17 of 30), and 63–99 % of the scored holdout rows of earlier regimes carried an index absent from the context. A difference near zero therefore does not show that the boundaries carry no information.

*Segment length.* Table 11 repeats the two main comparisons of Section 4.3 with segments of 1,000 and 5,000 rows. The sign is the same at all three lengths in every case except the CBRS dual memory against the class-balanced dual memory on Insects (abrupt) at 200 rows, where all three differences lie within ±0.51 pp; on Gas the differences grow with the segment length, which leaves only two segments at 5,000 rows. Scoring only checkpoints more than 5,000 rows after a segment's end, the registered lag analysis, gives the same signs as $\mathrm{RET}_{\mathrm{seg}}$ in both comparisons.

*Worst-retained regime (post hoc).* $\mathrm{RET}_{\mathrm{worst}}$, the lowest retention over the ended regimes of a run, was defined after the round-5 results and registered before round 6. The regime-balanced CBRS dual memory minus the CBRS dual memory gives +0.55 [−1.40, +2.49] (6/10) and +1.60 [+0.96, +2.24] (9/10) pp on Insects (abrupt), +12.97 [+7.87, +18.06] (10/10) and +5.49 [+1.95, +9.03] (9/10) pp on Gas and

**Table 11: Sensitivity to the segment length $L$ of $\mathrm{RET}_{\mathrm{seg}}$: paired differences in pp (mean over ten seeds) for keeping old rows (CBRS dual memory minus class-balanced dual memory) and for the allocation across regimes (regime-balanced CBRS dual memory minus CBRS dual memory), with segments of 1,000, 2,000 (main text) and 5,000 rows. On Gas, $L = 5{,}000$ leaves only two segments.**

| | | CBRS − class-bal. | | | Reg.-bal. CBRS − CBRS | | |
|---|---|---|---|---|---|---|---|
| Stream | Size | 1,000 | 2,000 | 5,000 | 1,000 | 2,000 | 5,000 |
| Insects (abrupt) | 200 | −0.05 | +0.10 | −0.51 | −1.23 | −1.24 | −1.08 |
| | 1000 | +1.22 | +1.28 | +1.32 | −0.91 | −0.96 | −0.94 |
| Gas | 200 | +10.07 | +11.22 | +13.80 | +1.11 | +1.84 | +2.71 |
| | 1000 | +9.17 | +11.04 | +16.98 | +0.52 | +0.87 | +1.26 |
| Electricity | 200 | +3.29 | +3.39 | +4.07 | +0.37 | +0.37 | +0.42 |
| | 1000 | +4.79 | +4.97 | +5.93 | −0.39 | −0.30 | −0.30 |
| Insects (gradual) | 200 | +4.78 | +4.92 | +5.62 | −1.73 | −1.74 | −2.25 |
| | 1000 | +5.39 | +5.47 | +6.11 | −0.24 | −0.23 | −0.23 |

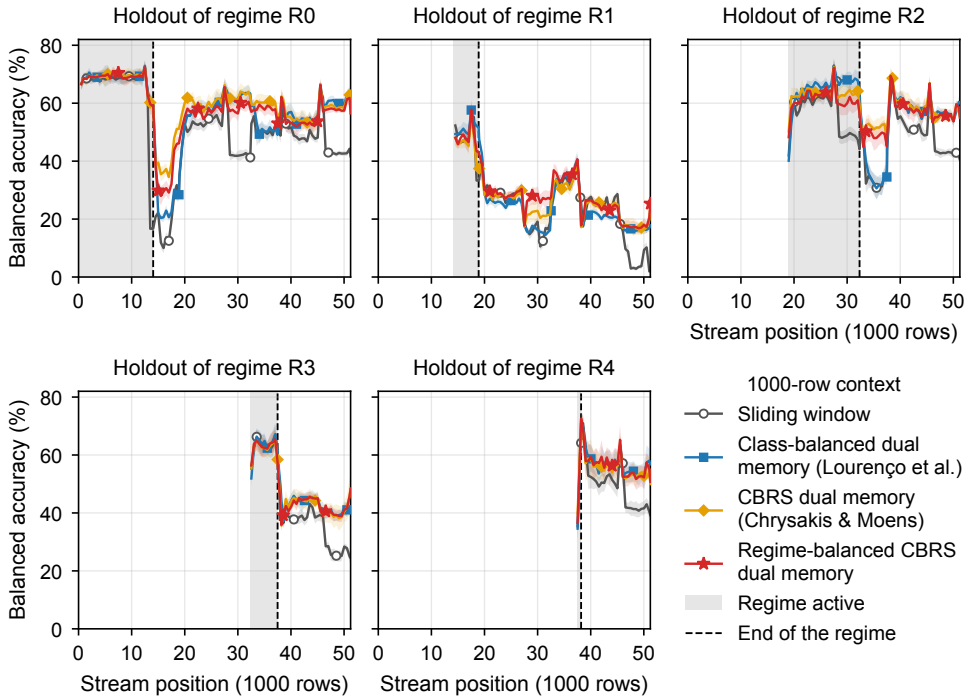



**Figure 11: Accuracy on the holdout set of each earlier regime as the Insects (abrupt) stream goes on (1000-row context; mean ± SD, ten seeds) for the four memories of the main text. Grey shading: the regime is active; dashed line: its end. After a regime ends, the sliding window loses much of it; at the end of the stream it lies below the three dual memories on every regime, on R1 even below chance (16.7 %), as R1 is almost unrelated to the other regimes (Figure 15).**

+3.08 [−0.66, +6.81] (8/10) and +1.01 [−0.45, +2.46] (7/10) pp on Electricity (200 and 1000 rows; 95 % $t$ intervals and positive seeds; Holm-significant on Insects (abrupt) at 1000 rows and on Gas at both budgets). On Insects (gradual) only R0 has ended, and the difference equals that in RET (−9.05 and −4.61 pp).

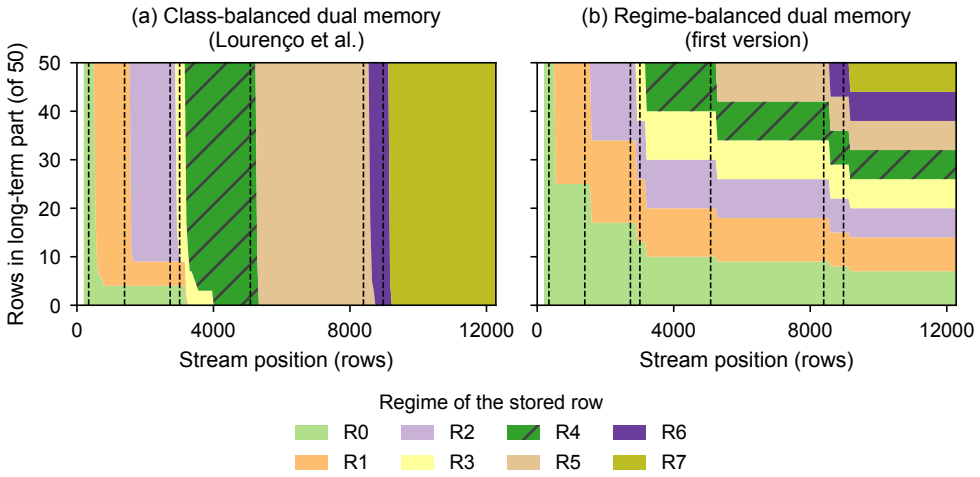



**Figure 12: What the 50-row long-term part holds over the Gas stream (200-row context; one run, seed 0). Colours: regime of each stored row; dashed lines: ends of regimes. (a) The class-balanced dual memory (Lourenço et al. [35]): after every change the long-term part fills with the newest regime, and at the end all 50 rows come from the last one, R7. (b) The regime-balanced dual memory (our first version): the long-term part is shared among all regimes seen so far (6–7 rows each at the end). Changes appear about 150 rows after a boundary, because new rows first pass through the short-term part. Over ten seeds, once a regime has ended, the whole 200-row class-balanced dual memory holds fewer than five of its rows during 83.6–97.9 % of the later updates on Gas (96.7–99.5 % on Electricity); the regime-balanced dual memory never does, by construction (equal shares per regime).**

*Other policies of round 2.* Besides the policies in the main text (the 200- and 1000-row windows and the 1000-row class-balanced dual memory and regime-balanced dual memory), round 2 compared the following (Figure 14; mean ADAPT / RET in %):

- sliding windows of 100, 300, 400, 600, 1500 and 2000 rows, from 73.70 / 33.81 (100 rows) to 75.51 / 42.11 (2000 rows), with the highest ADAPT at 600 rows (77.14);
- a class-balanced dual memory of 400 rows: 76.11 / 43.82;
- the boundary-free regime-balanced dual memory (ADWIN): 76.88 / 46.45;
- a class-balanced FIFO memory holding the 66 most recent rows of each class (400 rows): 67.97 / 46.47;
- the regime archive with all regimes: the 200-row window plus the archives of all finished regimes, 72.52 / 52.45, the highest retention but 4.3 pp less adaptation than the regime-balanced dual memory;
- the routed regime archive: the 200-row window plus the archive of the finished regime a query resembles most, with documented or ADWIN boundaries, 72.44 / 45.49 and 72.28 / 45.33.

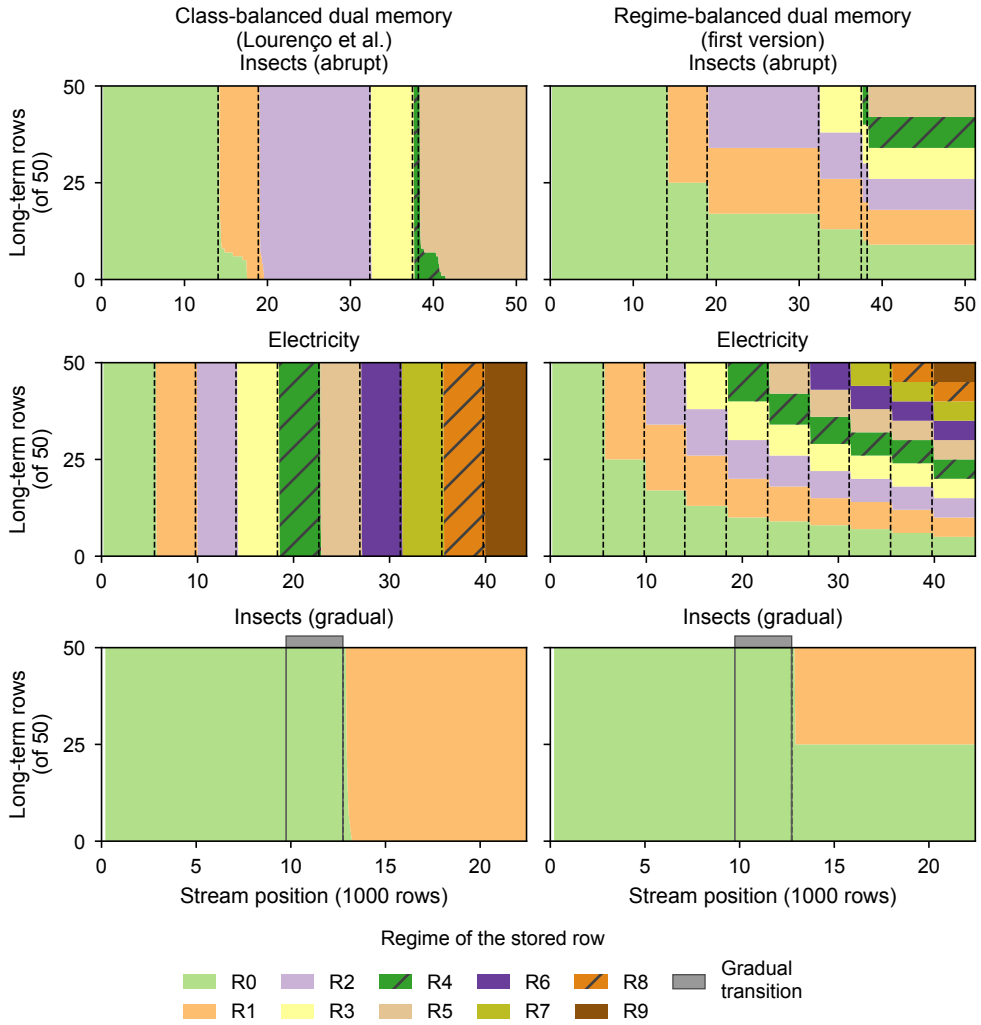



**Figure 13: What the 50-row long-term part holds over time on the other three streams (200-row context, one run, seed 0). Colours: regime of each stored row; dashed lines: regime ends; grey bar (Insects (gradual)): the gradual transition, which belongs to R0. The class-balanced dual memory (Lourenço et al. [35]) (left) keeps mainly the newest regime and at the end holds no older rows. The regime-balanced dual memory (our first version) (right) shares the long-term part evenly among all regimes seen so far. On Insects (abrupt), the class-balanced dual memory keeps a few old rows of some classes for a while after R0 and R4 end. On Insects (gradual) it replaces R0 within about 450 rows of the documented change point (about 2,900 rows with a 1000-row memory).**

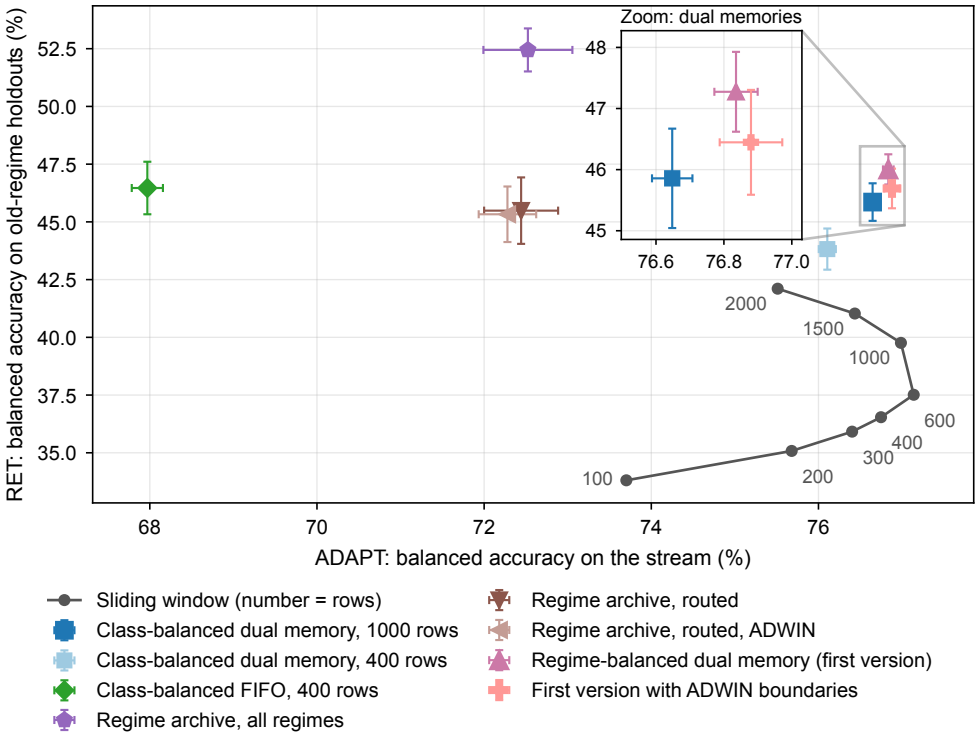



**Figure 14: Adaptation and retention of all 16 second-round policies on Insects (abrupt) (mean $\pm$ SD, ten seeds; SD of the sliding-window points $\leq 0.65$, not drawn). The grey line joins sliding windows of 100–2000 rows. Compare the 1000-row memories with the 1000-row window; the 200-row window was the pre-registered baseline (Section 3.5). The regime archives are the originally registered method (Appendix C). No policy is at least as good as the regime-balanced dual memory in both metrics. Inset: the three 1000-row dual memories.**

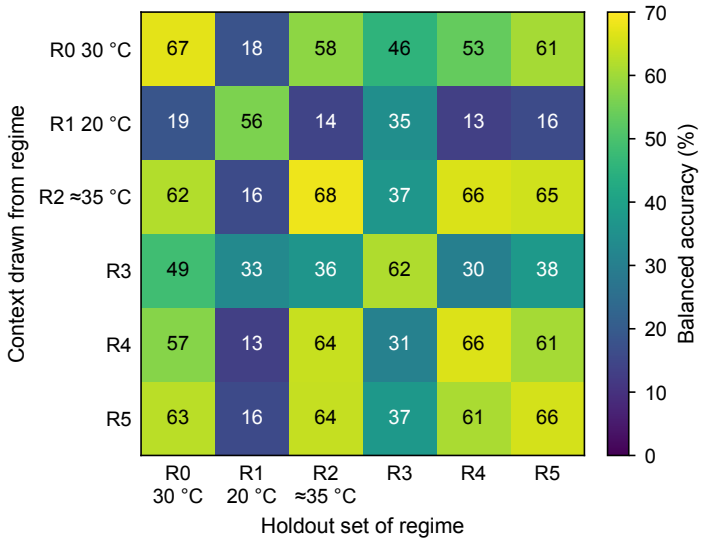



**Figure 15: How well a context from one regime predicts another (Insects (abrupt); balanced accuracy in %, mean over ten seeds). Each context holds 198 class-balanced rows drawn from one regime (rows); each column is the holdout set of one regime. The diagonal lies at 56–68 %; R1 is almost unrelated to the other regimes (13–35 % in its row and column), whereas R0, R2, R4 and R5 predict each other well (53–66 %). Forgetting R1 therefore cannot be made up by rows of other regimes.**

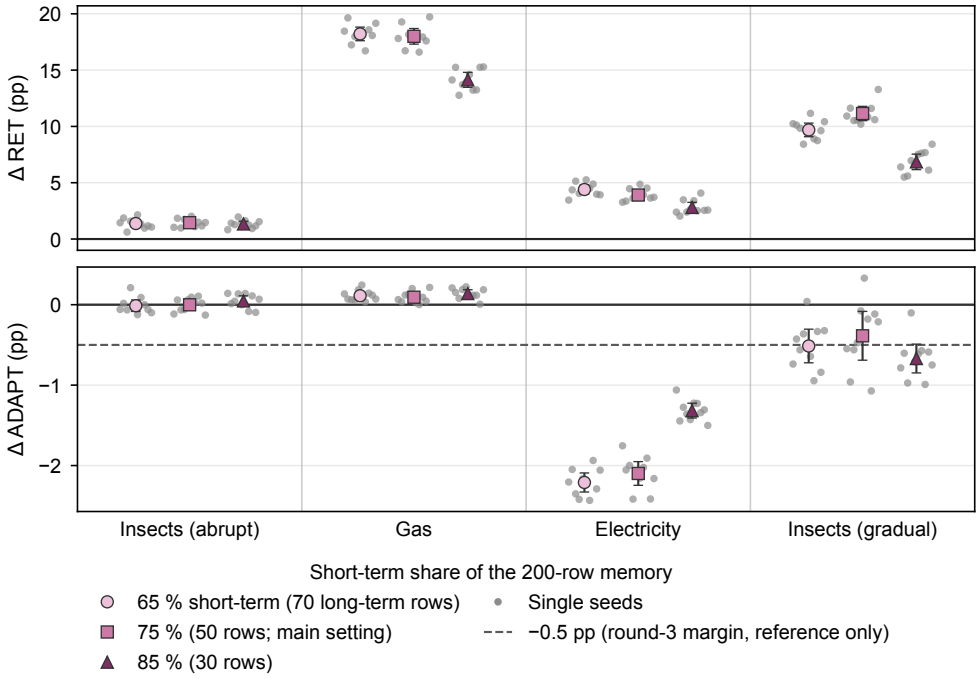



**Figure 16: Exploratory analysis: regime-balanced dual memory minus class-balanced dual memory (Lourenço et al. [35]) when the short-term part takes 65, 75 or 85 % of a 200-row memory (75 % is the main setting). Markers: mean over ten seeds with 95 % $t$ intervals; grey dots: single seeds. Top: the gain in official RET stays positive for every stream, share and seed; in $\text{RET}_{\text{seg}}$ it does so on the other three streams but not on Insects (abrupt) (Section 4.3). Bottom: the cost in ADAPT on Electricity stays beyond the $-0.5$ pp reference line at every share.**

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